Soft vs Rigid Gripper Benchmark for Cloth Grasping

A Force-Depth-Tilt Characterization of Parallel-Jaw and Finray Geometries

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Abstract

Robotic grasping of thin, deformable fabric remains an unsolved problem in industrial automation. Two competing gripper geometries are routinely proposed. Rigid parallel-jaw fingers offer precise force control but narrow operational tolerance, while compliant finray-effect fingers trade force precision for a wider operational envelope. In this work, an experimental evaluation of that trade is presented on the task of picking a single flat cloth off a surface, comparing two rigid parallel-jaw configurations and four 3D-printed finray configurations, six gripper variants in all, on a Franka Research 3 arm. Press depth and tilt angle were chosen as the evaluation axes because they are the placement errors that a deployed system cannot fully control. A low-accuracy or coarsely calibrated arm carries millimetre-scale surface-height error and degree-scale orientation error, and both of them must be absorbed by the gripper. Stationary contact force was the primary outcome and grasp success the secondary outcome.

Across 840 analysed trials, the rigid jaws held cloth only within a 1.5 mm depth window, or 2.5 mm with a compliant pad, bounded by a force safety limit. Under tilt these windows narrowed further, to between 0.5 and 2.5 mm, and two of the stock-jaw conditions produced no successful grasp at any depth. However, every finray variant kept the full 10 mm calibrated analysis window at every orientation tested, so the tilt did not reduce the finray working range at all. In addition, all four finray variants were found to share a single force-depth curve once each was normalised by its own peak force, with a mean pairwise error below 1%. The finger geometry therefore sets the scale of the contact force, although it does not change the shape of the response. It is worth noting that the contact force near the boundary of the working range still varies with the tilt direction, although this sensitivity disappears once the finger is

pressed past the buckling knee.

Abrégé

La saisie robotisée de tissus minces et déformables demeure un problème ouvert en automatisation industrielle. Deux géométries de préhenseur sont couramment proposées : les doigts rigides à mâchoires parallèles, qui offrent un contrôle de force précis mais une tolérance opérationnelle étroite, et les doigts souples de type finray, qui réduisent la précision de force pour une enveloppe opérationnelle plus large. [Cette thèse présente une évaluation expérimentale de ce compromis pour saisir un tissu plat posé sur une surface, en comparant deux configurations rigides à mâchoires parallèles et quatre configurations finray imprimées en 3D, soit six variantes de préhenseur, sur un bras Franka Research 3.] La profondeur d'appui et l'angle d'inclinaison ont été choisis comme axes d'évaluation parce qu'ils représentent les erreurs de placement qu'un système déployé ne peut pas totalement évaluer: un bras peu précis ou grossièrement calibré porte une erreur de hauteur de surface à l'échelle millimétrique et une erreur d'orientation à l'échelle du degré, que le préhenseur doit mitiger. La force de contact stationnaire a servi de variable de résultat principale et la réussite de la prise de résultat secondaire.

Sur 840 essais analysés, les mâchoires rigides n'ont tenu le tissu que dans une fenêtre de profondeur de 1,5 mm, ou de 2,5 mm avec un coussinet souple, bornée par une limite de sécurité en force. Sous inclinaison, ces fenêtres se sont encore rétrécies, entre 0,5 et 2,5 mm, et deux des conditions de la mâchoire d'origine n'ont produit aucune prise réussie à aucune profondeur. Cependant, chaque variante finray a conservé toute la fenêtre d'analyse calibrée de 10 mm à chaque orientation testée, de sorte que l'inclinaison n'a pas réduit la plage utile du finray. En outre, les quatre variantes finray suivent une seule courbe force-profondeur une fois normalisée par leur propre force maximale, avec une erreur moyenne par paire inférieure à 1%. La

géométrie du doigt fixe donc l'échelle de la force de contact, bien qu'elle ne change pas la forme de la réponse. Il est à noter que la force de contact près de la limite de la plage utile varie encore avec le sens de l'inclinaison, bien que cette sensibilité disparaîsse une fois le doigt appuyé au-delà du coude de flambement.

[TODO: Faire vérifier cette traduction par un collègue francophone avant le dépôt final (exigence de qualité, non de conformité : GPS exige seulement la présence d'un abrégé en français).]

Contribution to Original Knowledge

This thesis contributes the first controlled, parametric characterisation of how compliant finray-effect fingers and rigid parallel-jaw grippers grasp cloth as a function of press depth and gripper tilt, holding the robot, control policy, cloth specimen, and environment fixed and varying only the end-effector. Prior work on finray-family fingers has modelled or measured *lateral* grasping contact; no published model or benchmark, to the author's knowledge, characterises the axial tip-compression response: the rise–peak–valley–recovery force profile identified here, shared across four finray variants to within 1% normalised error. The principal empirical contribution is the demonstration that finray tilt robustness is *depth-gated*. The response is strongly tilt-sensitive near the no-contact boundary but robust to tilt once pressed past the buckling threshold, and the direction-dependent asymmetry about the finger's geometrically asymmetric axis likewise vanishes in the safe operating regime. This depth-dependent tilt robustness had not been quantitatively characterised in prior literature. Secondary contributions include a reproducible evaluation protocol and its open-source implementation, a documented per-variant tool-centre-point calibration procedure, and a geometric verification framework disentangling control-frame positioning error from physical gripper compliance under load.

Contribution of Authors

This thesis is the original work of the author, Shiyao Ni, under the supervision of Prof. Audrey Sedal. The author carried out all experimental design, hardware integration, software development, data collection, analysis, and writing.

The Franka FR3 manipulator was maintained by the MACRObotics Lab; the `fr3_pose_controller` ROS 2 control stack was developed and maintained by the McGill Applied Dynamics Lab. The 3D-printed finray fingers were fabricated using lab resources at the MACRObotics Lab. No portion of this thesis has been previously published.

This is a traditional (monograph-style) thesis; no chapter derives from a co-authored manuscript, and no collaborator contributed content requiring declaration under McGill GPS guidelines beyond the infrastructure acknowledgements above.

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Finally, I thank my family for their patience and encouragement.

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List of Abbreviations

API	Application Programming Interface
CAD	Computer-Aided Design
DOF	Degree(s) of Freedom
FCI	Franka Control Interface
FR3	Franka Research 3 (robot model)
GT	Ground Truth (calibrated surface height record)
NMSE	Normalised Mean-Squared Error
PEI	Polyetherimide (3D-printer build-plate surface)
RMSE	Root-Mean-Square Error
RGB-D	Red–Green–Blue plus Depth (camera modality)
ROS 2	Robot Operating System 2
RPY	Roll–Pitch–Yaw (orientation angles)
TCP	Tool Centre Point
TPU	Thermoplastic Polyurethane
URDF	Unified Robot Description Format

Notation

Frames and poses

$\mathcal{W}$, $\mathcal{F}$, $\mathcal{T}$	World (robot base), gripper flange, and gripper tip frames
$\mathbf{p}$	Position vector in $\mathbb{R}^3$ (boldface); $\mathbf{p}_{\mathcal{F}}$, $\mathbf{p}_{\mathcal{T}}$ denote the flange and tip positions expressed in $\mathcal{W}$
R, R_0	Orientation of the gripper as a rotation matrix, $R \in SO(3)$; $R_0 = R_x(180°)$ is the nominal down-pointing orientation
$R_x(\phi)$, $R_y(\phi)$	Elementary rotation matrix for a rotation by angle ϕ about the fixed world x or y axis
$\hat{\mathbf{z}}$	Unit vector along $+z$
z_{TCP}	Per-variant tool-centre-point offset from flange to tip along the flange's local z axis (mm); $z_{\mathrm{TCP}}^{(v)}$ denotes the calibrated value for variant v
x_0, y_0	Fixed probe/target position in the world xy plane (mm)

Design parameters (fixed per variant)

t	Finray finger blade thickness (mm) (Fig. 3.1)
α	Baseline seat-mount tilt angle, about the gripper's local y axis, introduced by the 3D-printed inclined seat (°) (Fig. 3.1)

Runtime and trial variables

θ_x, θ_y	Commanded robot-arm tilt angles about the world x / y axes (which coincide with the gripper's local x / y axis lines at R_0), applied as $R_{\text{tilt}} = R_y(\theta_y)\, R_x(\theta_x)$ (extrinsic) (°); distinct from, and never summed with, the seat angle α (Fig. 3.3)
d	Press depth: distance the gripper fingertip is driven below the detected cloth/table surface (mm)
d_{cal}	Calibrated press depth: $d - d_{\text{first success}}$, zeroed at each series' first majority-success cell (mm)
z_{surf}	Detected cloth/table surface height at the tip's xy location (mm)
a	Tilt axis identifier (x or y)
$b(d, a)$	Signed tilt bias: $b(d, a) = \lvert F_z^{+2.5°}(d, a)\rvert - \lvert F_z^{-2.5°}(d, a)\rvert$ (N)

Force measurement

$\mathbf{F}_{\text{ext}}$	Franka external wrench force vector at the end-effector flange (N); F_x, F_y, F_z denote its components
F_z^{bias}	Free-air F_z bias captured at session start (N)
$F_z^{(i)}$	The i-th stationary F_z sample within a trial (N)
N	Number of stationary force samples pooled in a (variant, depth, tilt) cell
$\bar{F}_z$, σ_{F_z}	Per-cell mean and standard deviation of F_z over pooled trials
$\sigma_{z,\text{tcp}}$	Across-tap standard deviation of the touch-off z measurement during TCP calibration (mm; distinct from σ_{F_z} above)

Derived quantities

L_{eff}	Effective finger lever-arm length inferred from the mount-tilt/depth-shift equivalence (mm)

Preface

This work grew out of a practical frustration. In the laboratory, as in much of the published literature, robotic cloth-manipulation demonstrations routinely begin with a human operator placing the fabric into the robot's gripper. Although the acquisition of the first grasp, that is, the picking of a thin, flat cloth lying on a surface, is the step that most limits autonomy, it is also the step that is most often skipped. This gap is usually approached with additional perception or with learning. However, a hardware question is asked in this work instead, and it is aimed to assess how much of the problem is removed when the gripper itself is compliant, and what is traded away in exchange.

All chapters of this thesis are original and previously unpublished work. Collaborative and infrastructural contributions are declared in the Contribution of Authors section.

1

Introduction

1.1 Motivation

Fabric-handling automation is a long-standing open problem in industrial robotics, with applications spanning apparel manufacturing and textile logistics, laundry processing and folding, hospital bedding management, assistive robotics for elderly or disabled individuals, and the packaging of soft goods. Yet, a striking fraction of published robot-manipulation demonstrations sidesteps the hardest step entirely. The object is handed to the robot's gripper by a human operator, and the demonstrated autonomy begins only after the grasp already exists.[1] For cloth specifically, acquiring the first grasp, picking a thin, flat fabric directly off a surface, remains the bottleneck that limits how autonomous these systems can truly be. Cloth is uniquely difficult to grasp robustly. It deforms continuously under contact, exhibits no fixed graspable feature, and has a near-zero bending stiffness that allows it to slip out of a poorly

[1] One prominent recent example is a widely circulated January 2024 video of the Tesla Optimus humanoid folding a shirt was subsequently acknowledged by its poster to be teleoperated, with the shirt pre-staged in a fixed setup [1, 2].

placed gripper. Although conventional rigid-jaw grippers are designed for sturdy objects with well-defined geometry, they perform poorly on cloth without elaborate tactile sensing or human-tuned approach trajectories.

A class of compliant grippers based on the finray effect, a passive bending mechanism inspired by fish-fin anatomy, has been proposed as a solution. Finray fingers passively wrap around the contact surface, increasing contact area and gripping force without requiring active force control. However, this compliance comes at the cost of force-modelling difficulty. The finger geometry deforms under load, the contact point shifts, and tilt-induced asymmetries arise that have no analogue in the rigid-jaw case.

To what extent does the finray's compliance trade single-axis force precision for multi-axis operational robustness? This is the central question addressed in this work. Specifically, it is aimed to assess how the finray's contact force responds to deviations in press depth and gripper tilt angle, which are the two design degrees of freedom most likely to be uncontrolled in real deployment.

1.2 Research Questions and Contributions

The following research questions are answered in this work through a controlled experimental evaluation on a Franka FR3 collaborative robot. In the evaluation, the gripper itself is treated as the experimental variable, while the robot, control policy, cloth, and environment are held identical across all conditions.

1. How does the contact force vary with press depth, for each finger design?
2. How much can the gripper be tilted before the grasp becomes unreliable?
3. Does tilt tolerance depend on the press depth?
4. What is the soft gripper's response under compression?

The contributions of this work are summarised as follows.

- A reproducible evaluation protocol for cloth-grasp force characterisation on a Franka FR3, with full source code published alongside this thesis.

- A two-dimensional depth–tilt experimental matrix sampling six gripper variants, namely two rigid parallel-jaw configurations and four finray prints spanning blade thickness, seat angle, and print instance.

- Empirical evidence of a depth-dependent tilt robustness transition in finray grippers, with quantitative characterisation of the transition depth.

- A documented calibration procedure for per-variant tool-centre-point (TCP) offset using differential force touch-off against a rigid reference.

- A geometric verification framework that disentangles control-frame positioning error from physical gripper compliance under load.

Scope. The evaluation deliberately isolates the *contact* phase of cloth grasping, the interval between first gripper–cloth contact and a completed lift, and characterises it over a continuous depth–tilt parameter space. Downstream task-level evaluation, meaning folding, placing, and dragging under the same policy, is future work (Chapter 7). This narrowing is intentional, since the contact phase is usually bundled inside end-to-end task pipelines, and its mechanics have therefore not previously been mapped as a function of gripper design.

1.3 Thesis Organisation

Chapter 2 reviews the literature on cloth grasping, compliant grippers, and force-controlled approach strategies. Chapter 3 describes the experimental hardware, the kinematic calibration procedure, and the data-acquisition pipeline. Chapter 4 presents the two experimental campaigns, which are the depth-sweep characterisation (Experiment 1) across all six gripper variants at zero tilt, the tilt-sweep characterisation (Experiment 2) at boundary and safe depths, and the planned inclined-surface pilot demonstration (Experiment 3). Chapter 5 reports the resulting force–depth–tilt characterisation. Chapter 6 synthesises the findings into a model of depth-dependent tilt robustness. Chapter 7 concludes and identifies directions for future work.

2

Background and Related Work

2.1 Cloth Grasping in Robotics

Robotic manipulation of cloth and other deformable fabric is difficult because the object being manipulated lacks a fixed geometry. Unlike a rigid part, a cloth's shape is itself part of the manipulation state. Although the geometry of a rigid part can be assumed from a physical model, the shape of a cloth must typically be inferred from vision. Early work in this space focused on perceiving that shape well enough to act on it. Kita et al. [3] fit a deformable mass-spring model of a held garment to stereo-camera silhouette observations, and use the best-matching simulated deformation to estimate the position of an unheld part of the garment for a second manipulator to grasp. Maitin-Shepard et al. [4] adopt a complementary, model-light approach in which a multiple-view geometric-cue detector locates cloth corners directly from silhouettes, in a manner robust to texture and colour variation, letting a general-purpose PR2 manipulator pick and fold towels of varying material and appearance without a specialised end-effector. Miller et al. [5] generalise this line of work with a polygonal-approximation geometric model that tracks a garment's silhouette through an entire folding sequence, allowing one parameterised folding strategy to transfer across previously unseen towels, pants, and shirts. Li et al. [6] employ a predictive thin-shell simulation fit to RGB-D observations to plan regrasp actions that unfold and flatten a crumpled garment, selecting the grasp points expected to maximise resulting flatness. In all of these results, cloth state estimation is made tractable by a physics-informed or geometric shape prior, and not by raw image features.

More recent work has moved from hand-designed geometric priors toward learned policies and richer sensing. Saxena and Shibata [7] train convolutional classifiers to recognise garment type and locate hem/waistline grasp points directly from images, as the perception front-end of a robotic dressing-assistance pipeline. Wu et al. [8] show that a model-free visual reinforcement-learning policy for cloth and rope manipulation can be learned an order of magnitude faster than a jointly-learned pick-and-place policy, by conditioning only on the *placing* action and selecting picks that maximise the value of the resulting placement, entirely without human demonstrations. Tirumala et al. [9] mount a tactile sensor on a Franka gripper fingertip and train a classifier on the resulting tactile signal to determine how many layers of cloth have been grasped, re-grasping until exactly one or two layers are singulated from a stack. Longhini et al. [10] survey the field beyond these representative examples, spanning perception, dynamics modelling, benchmarking, and control specifically for textile manipulation. The tactile result is direct evidence that certain cloth-grasp ambiguities cannot be resolved by vision alone. However, the force signal is used there to classify a grasp that has already been made, and how contact force develops while the grasp is being acquired still remains unclear.

2.2 Strategies for Acquiring the First Grasp

Because a flat cloth on a surface presents no protruding feature for a conventional pinch grasp, two broad families of workaround have been developed in the literature. The first manipulates the *environment or cloth state* until a graspable feature exists. The second replaces the *end-effector* with a mechanism that does not require one.

2.2.1 Environment-Assisted and State-Changing Actions

The first family keeps a standard gripper and instead spends actions creating a graspable feature before the pick: a fold, a wrinkle, or a free edge. Ramisa et al. [11] detect and rank wrinkles in depth images of crumpled laundry, directing a standard gripper to the most graspable bump on the cloth instead of to a flat region. Qian et al. [12] segment cloth edges and corners, as distinct from wrinkles and folds, in depth images and couple the segmentation to a sliding grasp policy that approaches along the estimated grasp direction. At the dynamic extreme, Ha and Song's FlingBot [13] shows that a high-velocity fling primitive can unfold a crumpled cloth in a few actions. There the environment interaction resets the cloth to a flatter, more predictable state, and no pinchable feature is created. These strategies are effective, but each spends additional actions and sensing before the grasp itself. In addition, their reliability degrades in the flat-cloth-on-surface configuration targeted in this work, where there is no wrinkle to find and no free edge to slide against.

2.2.2 Specialised End-Effectors

The second family redesigns the gripper so that a protruding feature is unnecessary. Monkman's survey of grippers for fibrous materials [14] catalogues the classical industrial approaches, developed largely for garment manufacturing: intrusive (needle/pin) grippers that pierce the top ply, and adhesive or electrostatic surface grippers. Electroadhesion has since matured substantially; Guo et al. [15] review its use across robotics and note its suitability for delicate, flexible materials at low energy cost. Non-contact airflow methods lift fabric without touching it at all; Ozcelik and Erzincanli's Bernoulli-principle end-effector [16] suspends a garment ply on a radial airflow film. More recently, variable-friction fingertips have blurred the line between gripper and manipulation policy. Jiang et al. [17] switch fingertip surfaces between high- and low-friction states to slide, pinch, and unfold fabric in-hand. These end-effectors trade generality for capability. Needles risk damaging fine textiles, and electroadhesion and airflow devices require dedicated power or pneumatic infrastructure and are largely single-purpose. Although each of them is effective within its own material class, all of them abandon the ubiquitous two-jaw form factor provided by general-purpose robot platforms, the Franka Hand used in this work among them.

2.2.3 The Gap: the Gripper as an Experimental Variable

Across both families, the gripper hardware is treated as *fixed*, and the innovation budget is spent on perception, policy, or a task-specific mechanism. Direct comparisons between compliant and rigid jaw-type grippers do exist. Li et al. [18] systematically review soft–rigid hybrid grippers against industrial parallel-jaw grippers across payload, adaptability, and precision metrics. However, such comparisons are conducted on rigid or quasi-rigid workpieces and not on cloth, and they report capability envelopes, not parametric contact behaviour.

The nearest precedent is the household-clothing benchmark of Clark et al. [19]. Several commercial end-effectors are evaluated there on four cloth-specific benchmarks: edge-drag accuracy, edge-grasp resilience, crumpled-object encapsulation, and fold generation. The end-effectors included finray-style compliant fingers mounted on a Franka Emika Hand, the standard Franka Hand, a Robotiq 2F-140, and the OpenHand Model T42. Their results establish empirically that end-effector selection materially changes cloth-manipulation outcomes, and finray fingers are evaluated there on the same robot platform used in this work. The evaluation is scored at the *task* level, and each benchmark reports the success or accuracy of a completed manoeuvre with the approach pose fixed per task. Although end-effector selection is shown to matter, how the grasp itself degrades as the approach conditions vary is not measured, and the continuous dependence of contact force and grasp outcome on press depth and gripper tilt, which any deployed system experiences as positioning error, still remains unclear.

Therefore, none of the work above addresses the specific question investigated in this work. Grasp *success* is treated in the cloth-manipulation literature as an outcome to be predicted or achieved at a fixed operating point. It is not treated as a function of a continuous physical parameter space of press depth and gripper tilt, whose boundary can be mapped and compared across gripper geometries. The evaluation reported here is deliberately narrower in manipulation scope than a folding or dressing pipeline, and correspondingly more quantitative about the mechanics of the grasp itself. The two-jaw form factor is kept fixed, only the finger design (t, α) (defined in Figure 3.1) is varied, and the resulting contact-force landscape is mapped under the positional and angular uncertainty (d, θ_x, θ_y) that any deployed system must tolerate.

2.3 Compliant Gripper Designs

The finray-effect finger design used in this thesis originates in a patent by Kniese [20], who observed that a structural element built from two flexible longitudinal members joined by hinged cross-struts deflects *toward* an applied lateral load rather than away from it, the counter-intuitive "bends toward the pressure" behaviour later commercialised by Festo as the Fin Ray Effect®. Festo's own current technical documentation for its DHAS gripper-finger product line [21] describes this structure explicitly as two flexible bands joined by cross-ribs at flex hinges, and reports measured retention-force, lateral-force, and indentation-depth curves as functions of gripping force and workpiece diameter. The depth-sweep experiments of this thesis characterise the same three quantities, force, indentation depth, and contact geometry, directly on 3D-printed finray fingers rather than the commercial product.

Academic characterisation of the finray design has followed two directions: analytic modelling and empirical/manufacturing studies. Shan and Birglen [22] developed an analytic mechanical model of finray fingers to predict grasp force and finger deformation from geometric parameters, addressing the absence of a convenient design tool for this class of compliant finger. Their model, like the empirical studies that follow it, addresses *lateral* grasping contact; no published model, to the author's knowledge, predicts the axial tip-compression response, the rise–peak–valley–recovery profile characterised empirically in this thesis. On the empirical side, Crooks et al. [23] modified the finray design by angling its internal crossbeams to introduce a directional bending bias, fabricated by multi-material 3D printing, and reported an approximately 40% increase in holding capacity over a traditional finray finger from this geometric change alone. Elgeneidy et al. [24] characterised fully 3D-printed finray fingers made from flexible thermoplastic polyurethane (TPU) filament, varying crossbeam rib angle and spacing, and found that at large tip displacements the ribs contact one another (*layer jamming*), substantially increasing force generation and producing a tunable-stiffness effect. Lang et al. [25] 3D-printed finray fingers in both TPU 60A and TPU 95A and tensile-tested them: the softer 60A durometer gave the better force/compliance balance and left delicate fruit undamaged, while 95A reached substantially higher forces but visibly bruised the fruit. That result is prior evidence that finger material hardness, independent of finger geometry, is a first-order design lever for a compliant gripper; the manufacturing-variability results of Section 5.1 are consistent with it.

The operational-envelope argument for compliant hands, that compliance buys tolerance to positioning error, was established quantitatively by Dollar and Howe's SDM Hand [26]: an underactuated hand with compliant polymer fingers that grasped a 48 mm cylinder at positions offset by more than 100% of the object size in each direction, while keeping acquisition contact forces low, where a rigid gripper would require millimetre-level placement. Their successful-grasp-range maps over an xy offset grid are the direct methodological ancestor of the depth–tilt working-window maps produced in this thesis, transposed from rigid household objects to cloth and from lateral offset to press depth and tilt. Teeple et al. [27] decomposed the benefit of a pneumatic soft gripper's compliance into per-axis contributions when handling thin, flexible materials: vertical compliance was shown to buy robustness to large vertical positioning uncertainty, measured head-to-head against a rigid gripper, while lateral and rotational compliance lengthen the response window and cap the tensile force during snags. Their vertical-uncertainty comparison is the single-gripper-pair analogue of this thesis's central question; the evaluation here extends it across a family of finger designs, adds the tilt dimension, and grounds it on cloth with a fully passive compliant finger.

Compliant gripping is not achieved solely through finray-style flexure. Granular jamming is the other major compliant-gripping paradigm in the literature: Brown et al. [28] demonstrated that a latex membrane filled with granular material conforms to an object's shape under positive pressure and rigidifies under vacuum, enabling passive universal grasping without dedicated fingers at all, and Amend et al. [29] extended this with a combined positive/negative pressure actuation cycle for faster, more reliable grasp-and-release cycles. A third paradigm, pneumatic elastomer actuation, underlies commercial soft grippers such as those from Soft Robotics Inc.; the technology traces to pneumatic-network (PneuNet) elastomer actuators developed by Ilievski et al. [30], and the commercial mGrip product line [31] demonstrates that compliant gripping has moved beyond academic prototypes into food-handling industrial deployment. This thesis's finray variants and rigid parallel-jaw comparator sit within this broader design space as one specific, geometrically simple point in it, chosen because a finray finger, unlike a jamming or pneumatic design, requires no additional actuation hardware beyond the gripper's own jaw motion.

2.4 Force-Controlled Grasping

The theoretical basis for treating contact force, rather than position alone, as a control variable during manipulation is Hogan's impedance control framework [32, 33, 34]: neither pure position control nor pure force control is adequate for a task involving contact with an environment of unknown or varying mechanical properties, so the manipulator should instead regulate the dynamic relationship between its motion and the interaction force at its end-point, the impedance. This thesis's trial procedure, which presses the gripper to a commanded depth and then reports the resulting *stationary* contact force F_z rather than commanding force directly, is an impedance-control-style measurement: depth is the commanded quantity, force is the observed response, and the relationship between them (Chapter 5) is the impedance curve Hogan's framework predicts should characterise contact with an unknown environment, here the cloth-covered rigid table.

On the Franka FR3 platform specifically, libfranka exposes two real-time command modes: external torque-level control, for which a Cartesian impedance controller is distributed as a reference implementation, and Cartesian pose motion generation, in which the client streams pose targets at the 1 kHz servo rate and tracking is delegated to the arm's internal impedance controller [35]. The tilt-aware pose targeting used in this thesis's trial procedure (Section 3.4) uses the latter mode. The in-house `fr3_pose_controller` commands Cartesian poses through libfranka's pose interface, so the contact interaction is mediated by the robot's internal impedance law rather than a user-tuned stiffness.

Contact force in this thesis is measured through the Franka FCI's own external-wrench estimate; a second, independent channel using an array of Tekscan FlexiForce resistive force sensors was provisioned, but a repeatable calibration could not be established in time and the cross-check is carried as future work (Chapter 7). FlexiForce is a thin-film piezoresistive sensor marketed explicitly for grip-force measurement in robotic and surgical applications [36], and has prior academic precedent for this use: Sharan and Onwubolu [37] instrumented a two-finger pick-and-place gripper with a FlexiForce sensor to measure grip force directly against known payload weight. An alternative tactile-sensing approach not used in this thesis but relevant for future work is vision-based tactile sensing: Yuan et al.'s GelSight [38] sensor measures the deformation of a soft elastomer contact surface via an internal camera, recovering high-resolution contact geometry from which force and slip can be inferred, in contrast to the direct but lower-resolution force read-out of FlexiForce.

Finally, the trial procedure's touch-off step, in which the gripper descends under position control until a chosen contact criterion is met before switching to the pressed-depth grasp, follows a long lineage of contact-detection strategies in manipulation. Whitney's classical analysis of quasi-static assembly [39] derives the geometric and force-equilibrium conditions under which measured contact force can be used to detect and guide compliant insertion, the analytical basis for the Remote Center Compliance device and, more generally, for using force as a contact-detection signal rather than only as an outcome measure. De Luca et al. [40] showed that contact can be detected from proprioceptive joint measurements alone, without a dedicated force/torque sensor, via a generalised-momentum residual, a capability this thesis's FCI-based touch-off implicitly relies on when it uses the controller's own external-wrench estimate rather than an external load cell.

2.5 The Franka FR3 Platform

The Franka Research 3 (FR3) is a 7-degree-of-freedom, torque-controlled collaborative arm with seven integrated joint torque sensors, a 3 kg payload, and position repeatability under ± 0.1 mm, accessed in this thesis through the Franka Control Interface (FCI) [41, 42], a real-time control layer providing torque, position, and velocity access in both joint and Cartesian space at up to 1 kHz. This thesis controls the arm through `franka_ros2` [43], the official ROS 2 integration built on `libfranka` and `ros2_control`.

The FR3's predecessor, the Franka Emika Panda, is documented by its manufacturer as a widely adopted reference platform for robotics research and education [44], and a more recent survey by the same author quantifies its continued adoption as a de facto standard platform across manipulation, control, and human-robot-interaction research [45]. That widespread adoption has produced independent characterisation studies directly relevant to this thesis's instrumentation choices. Petrea et al. [46] empirically evaluated the accuracy of the Panda's built-in filtered external-force estimate against ground-truth load-cell measurements during controlled contact. The signal they evaluated is the same FCI external-wrench estimate used in this thesis as the Contact Force channel; their accuracy findings therefore bound how much that channel can be trusted, and motivate the independent force-sensor cross-check identified as future work. Gaz et al. [47] separately derived a full dynamic (inertial and friction) parameter model of the Panda's arm via penalty-based optimisation, a model since reused widely enough to have become a standard dynamic model of the platform. Beyond force sensing specifically, the Franka Panda/FR3's broad adoption as a manipulation-research testbed is illustrated by work such as Chi et al.'s [48] diffusion-policy visuomotor control framework, evaluated on contact-rich real-world manipulation tasks using a Franka Panda as one of its physical platforms. This adoption is directly relevant to the control mode chosen here: learned visuomotor policies of this kind emit streams of end-effector pose targets, the same command interface through which this thesis's trial procedure drives the arm (Section 3.4). The evaluation is therefore conducted through the interface a learned or simulation-trained policy would use in deployment, with sim-to-real transfer as the motivating trajectory. A compliant finger that tolerates centimetre-scale press-depth error and degree-scale tilt error widens the placement-error budget available to a policy trained in simulation, and the depth–tilt maps produced in this thesis quantify that budget as a function of gripper design.

2.6 Position and Summary

This thesis sits at the intersection of three threads: cloth-handling system design, finray-effect gripper characterisation, and force-controlled manipulation on collaborative robots. Prior work has established each thread independently. Cloth manipulation research has largely optimised for task-level success in folding, singulating, and dressing under vision-driven or learned control; compliant-gripper research has characterised finray and jamming designs primarily through force/holding-capacity metrics on rigid or quasi-rigid test objects; and impedance-control theory and its Franka FCI implementation provide the tools to regulate and measure contact force but have not, in the reviewed literature, been used to map how that force varies over a continuous depth–tilt parameter space for a fragile, non-rigid workpiece. The contribution of this thesis is a unified experimental treatment of all three on a single platform with consistent instrumentation, enabling direct quantitative comparison between rigid and compliant gripper response under deliberately perturbed conditions.

3

Experimental Setup and Methodology

3.1 Hardware Configuration

3.1.1 Robot Manipulator

A Franka Research 3 (FR3) 7-DOF cobot was used as the manipulator throughout this work. The arm was operated through the Franka Control Interface (FCI) using `libfranka 0.18.1` and the ROS 2 Humble control stack. Cartesian pose targets were streamed through libfranka's motion-generator interface by a custom `fr3_pose_controller` developed and maintained by the McGill Applied Dynamics Lab, with tracking delegated to the arm's internal impedance controller (Section 2.5). The controller accepts a sequence of (pose, twist) waypoints with associated time stamps and interpolates between them using a quintic polynomial; for the in-place re-orientation and the final precision descent phases, a cosine half-cycle S-curve interpolator (the "sine descent") is used to ensure zero start- and end-state velocities.

3.1.2 Gripper Variants Tested

Table 3.1: Six gripper variants evaluated, indexed by blade thickness t and baseline seat-mount tilt α (Figure 3.1). All finray variants share the same 3D-printed inclined-seat mount; only α changes between the $+2.5°$ baseline and the $+5.0°$ variant. The parallel-jaw variants do not use an inclined seat and have no defined t or α.

#	Variant	Type	Finger	t	α
1	`parallel_jaw_stock`	Rigid	Metal jaws (Franka Hand)	—	—
2	`parallel_jaw_TPU`	Rigid + pad	2.0 mm TPU 90A pads	—	—
3	`finray_18`	Compliant	TPU 90A print	18 mm	$+2.5°$
4	`finray_18_model2`	Compliant	Second print of #3	18 mm	$+2.5°$
5	`finray_26`	Compliant	TPU 90A print	26 mm	$+2.5°$
6	`finray_26_5deg`	Compliant	TPU 90A print, 5.0° seat	26 mm	$+5.0°$

The finray fingers and the parallel-jaw compliant pad were 3D-printed from Bambu Lab TPU 90A (Shore-A hardness 90) on a Bambu Lab X1 Carbon printer using the vendor-default TPU 90A profile in Bambu Studio (0.4 mm nozzle, default layer height, default infill, PEI-textured build plate).

Baseline inclined-seat mount. We define α as the baseline seat-mount tilt angle, annotated in the front view of Figure 3.1: the fixed rotation about the gripper's local y axis, the gripping direction, introduced by the 3D-printed seat adapter that carries every finray finger. The figure shows the mounting stack (flange → seat → finger) and the α and t design-parameter convention. All finray variants were mounted on the Franka Hand via this inclined seat; the seat was present in *every* finray variant tested in this thesis unless otherwise noted, and is the physical realisation of α. Variants whose name contains `_5deg` used an $\alpha = +5.0°$ seat in place of the default $\alpha = +2.5°$ seat; all other finray variants (`finray_18`, `finray_18_model2`, `finray_26`) used the $\alpha = +2.5°$ baseline; the parallel-jaw variants have no seat and no defined α. The $\alpha = +2.5°$ baseline was chosen so that the two opposing finray fingers converge inward toward each other rather than splaying apart. This closes the residual slit that would otherwise remain between the finger tips when the gripper is fully closed, and ensures firm contact along the fingers' full length during a successful grasp.

Choice of design-parameter values. The two blade thicknesses and the second seat angle were not arbitrary. The 26 mm blade matches the finger dimension of the handheld Universal Manipulation Interface (UMI) gripper [49], whose finray-type soft fingers have become a de facto reference design for learned manipulation, so results at $t = 26$ mm transfer directly to that widely used hardware. The 18 mm blade matches the thickness of the stock Franka parallel jaw, giving a compliant finger and a rigid finger of equal frontal dimension for the cross-class comparison. The $\alpha = +5.0°$ seat was chosen to sit outside the $\pm 2.5°$ runtime tilt range: with a seat at the baseline $+2.5°$, a runtime tilt of $-2.5°$ would exactly cancel the seat inclination and confound the two effects, whereas the 5.0° seat keeps the fixed mount contribution distinct from every runtime tilt tested.

Consequence for reported "0°" cells. Cells reported in Chapter 5 as 0° tilt denote that the robot arm is perpendicular to the surface ($\theta_x = \theta_y = 0$). Runtime tilt values ($\pm 2.5°$ about x or y) tilt the arm away from this perpendicular pose; the runtime tilt angles themselves are illustrated in Figure 3.3.

3.1.3 Cloth Specimen

All trials used a single cloth specimen: a 30×22.5 cm piece of single-layer plain-woven cotton (Figure 3.2). The same physical specimen was reused for the entire campaign and reset to a flat, wrinkle-free configuration before every trial (Section 3.6, step 0), so that each trial began from the flat-on-surface configuration this thesis targets.

3.1.4 Test Surface

The cloth rested on a rigid, flat acrylic plate that was the test surface for the entire campaign, fixed to the workbench so that its position and orientation were common to all variants and sessions. The surface height entering every trial was measured rather than assumed: the force touch-off of Section 3.2.2 re-detected the surface at each calibration, so the plate's absolute height was a per-session measured quantity, not a fixed constant.

[TODO: insert photograph of the test bench in situ and short caption describing the table material, flatness, and cloth-edge orientation relative to the gripper y axis.]

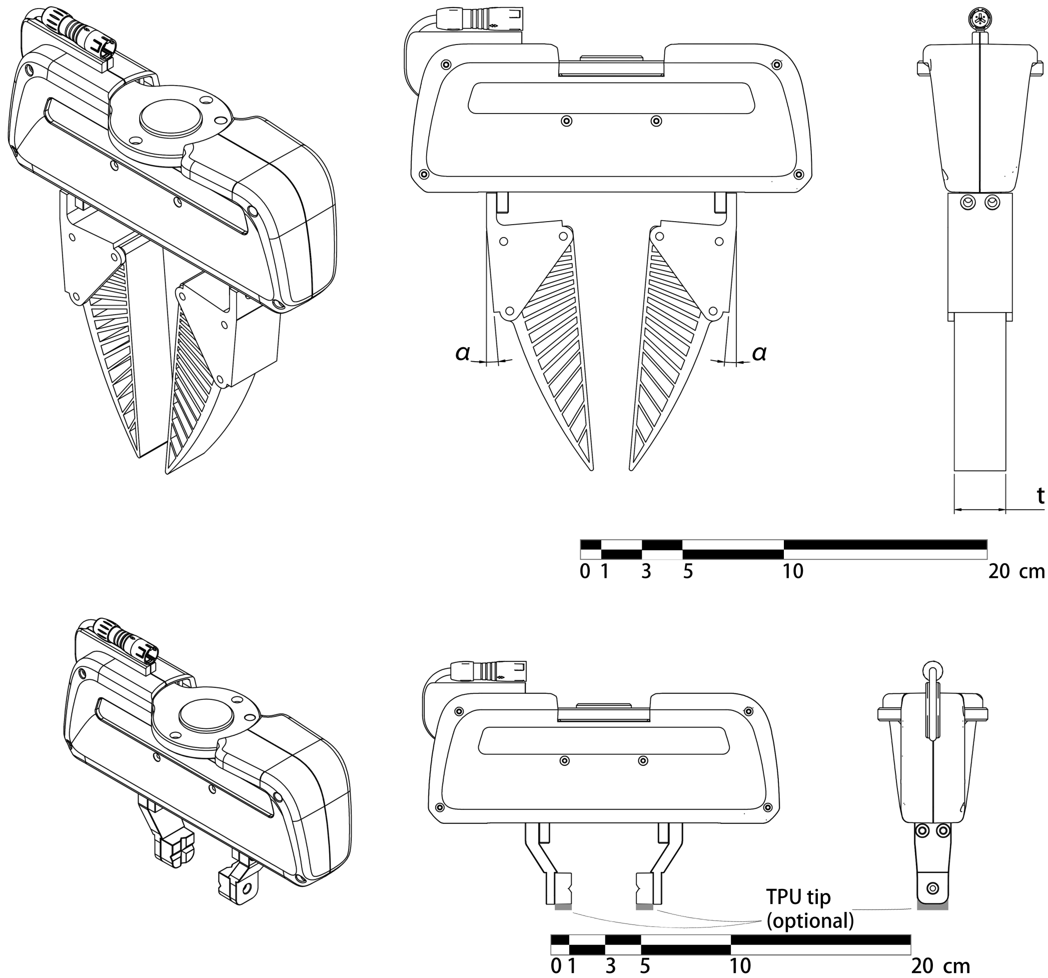


Figure 3.1: Both grippers in isometric, front, and side orthographic views, mounted on the Franka Hand. Top: a finray variant; the front view (centre) annotates the baseline seat-mount tilt α at each finger root, and the side view (right) annotates the finger blade thickness t. Bottom: the rigid parallel-jaw gripper with the optional TPU tip indicated. Scale bars in cm.

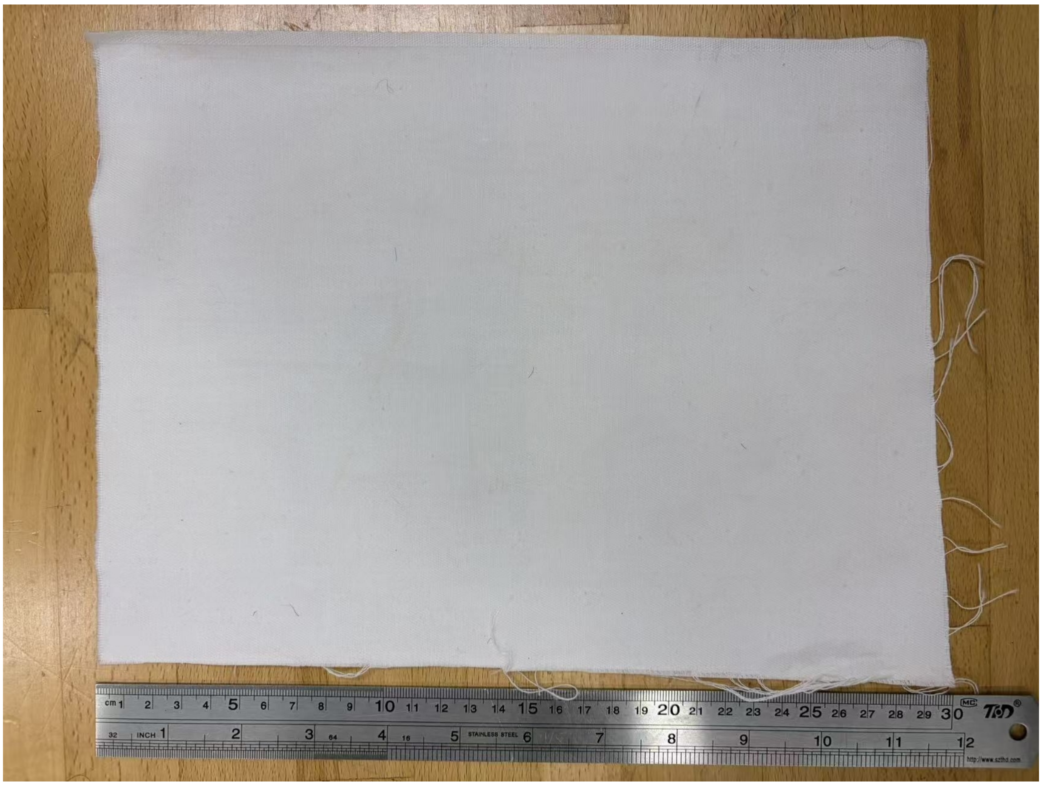


Figure 3.2: The single-layer plain-woven cotton specimen used throughout this thesis, 30×22.5 cm, shown against a ruler for scale.

3.2 Reference Frame and Tool-Centre Calibration

3.2.1 Frame Convention

Three frames are relevant: the world (robot base) frame $\mathcal{W}$, the gripper flange frame $\mathcal{F}$, and the gripper tip frame $\mathcal{T}$. The Franka URDF defines $\mathcal{F}$ at the mechanical hand mounting flange; the Franka API `robot.end_effector_pose` reports the pose of $\mathcal{F}$ in $\mathcal{W}$. The tip frame $\mathcal{T}$ is located along the local $+z$ axis of $\mathcal{F}$ by a per-variant offset z_{TCP}, which depends on the physical gripper (finger length, mounting fixtures, pads).

Throughout, $\mathbf{p} \in \mathbb{R}^3$ (boldface) denotes a position vector expressed in $\mathcal{W}$ unless otherwise noted, and $R_x(\phi)$, $R_y(\phi)$ denote the elementary rotation matrices for a right-handed rotation by angle ϕ about the fixed world x or y axis, so that a general gripper orientation is a composition $R \in SO(3)$ of these primitives. For a gripper at orientation R, the world-frame relationship between the tip and flange positions is

$$\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} + R \begin{bmatrix} 0 \\ 0 \\ z_{\text{TCP}} \end{bmatrix}. \tag{3.1}$$

For the rigid-down-pointing nominal orientation $R_0 = R_x(180°)$, the tip lies directly below the flange and $\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} - z_{\text{TCP}}\hat{\mathbf{z}}$.

The gripper-frame convention adopted here is: the gripping axis (along which the fingers close) is $\mathcal{F}$'s local y axis; the gripper is mirror-symmetric about the xz plane; consequently, rotation of the gripper about the local y axis is rotation about the geometrically asymmetric axis, while rotation about x is rotation about the symmetric axis. Because the nominal orientation $R_0 = R_x(180°)$ is itself a rotation about the world x axis, the world x and y axis *lines* coincide with the flange's local symmetric and gripping axis lines at R_0 (up to sign). The runtime tilt angles θ_x, θ_y introduced in Section 3.4 (Figure 3.3) are defined as rotations about these fixed world axes, and because the axis lines coincide with the gripper-local axes at R_0, the same symmetric-axis / gripping-axis language applies without ambiguity to θ_x and θ_y respectively.

3.2.2 TCP Offset Calibration

The z_{TCP} value differs by variant because different fingers attach to the same flange. To calibrate, each variant was driven to a common probe position (x_0, y_0) on a known rigid table and a force-feedback touch-off was performed: the gripper descended slowly in z until the external z force exceeded 3 N in magnitude, debounced for three consecutive 50 Hz samples. The flange z at the latch moment was recorded as $z_{\text{TCP}}^{\text{measured}}$.

Taking the rigid `parallel_jaw_stock` variant as the URDF reference ($z_{\text{TCP}} \equiv 0$ by definition), the per-variant TCP offset is

$$z_{\text{TCP}}^{(v)} = z_{\text{TCP}}^{(v),\text{measured}} - z_{\text{TCP}}^{\text{stock,measured}}. \tag{3.2}$$

Table 3.2 summarises the calibrated values used throughout this work.

Table 3.2: Calibrated per-variant TCP offsets from differential touch-off against the stock parallel jaw. Offsets are computed from the unrounded calibration records; displayed values are rounded independently. `finray_18_model2` is absent because it was re-calibrated on mounting (see text).

Variant	$z_{\text{TCP}}^{\text{measured}}$ (mm)	$z_{\text{TCP}}^{(v)}$ (mm)
`parallel_jaw_stock`	37.871	0.000
`parallel_jaw_TPU`	39.916	2.044
`finray_18`	131.842	93.971
`finray_26`	132.782	94.911
`finray_26_5deg`	130.094	92.223

The reproducibility of the touch-off was measured as the across-tap standard deviation $\sigma_{z,\text{tcp}}$ in mm; this quantity is distinct from the per-cell force standard deviation σ_{F_z} of Section 3.7. For `finray_26_5deg`, $\sigma_{z,\text{tcp}} = 183\,\mu\text{m}$ owing to a wobble component introduced by the inclined seat; the other four variants had $\sigma_{z,\text{tcp}} \leq 102\,\mu\text{m}$. The second print (`finray_18_model2`) shares the finger geometry and seat of `finray_18`; its surface reference and TCP offset were re-established by the same touch-off procedure when it was mounted, rather than reusing the first print's calibration.

3.3 Force Measurement

External forces $\mathbf{F}_{\text{ext}}$ at the flange were estimated by the Franka internal model from joint torques, available via the `libfranka` RobotState API at 1 kHz and subsampled to 30 Hz for session logging. A free-air bias capture was performed at the start of every session: the gripper was held at a known pose 30–80 mm above the cloth surface and 30 samples were averaged over 1 s to provide F_z^{bias}. This bias was recorded in each session JSON but *not* subtracted from the raw force time series at acquisition time; downstream analyses subtract it where appropriate.

3.4 Tilt-Aware Pose Targeting

When the gripper is tilted, the flange position required to place the tip at a fixed probe location $(x_0, y_0, z_{\text{surf}})$ shifts laterally. From equation (3.1), the required flange position is

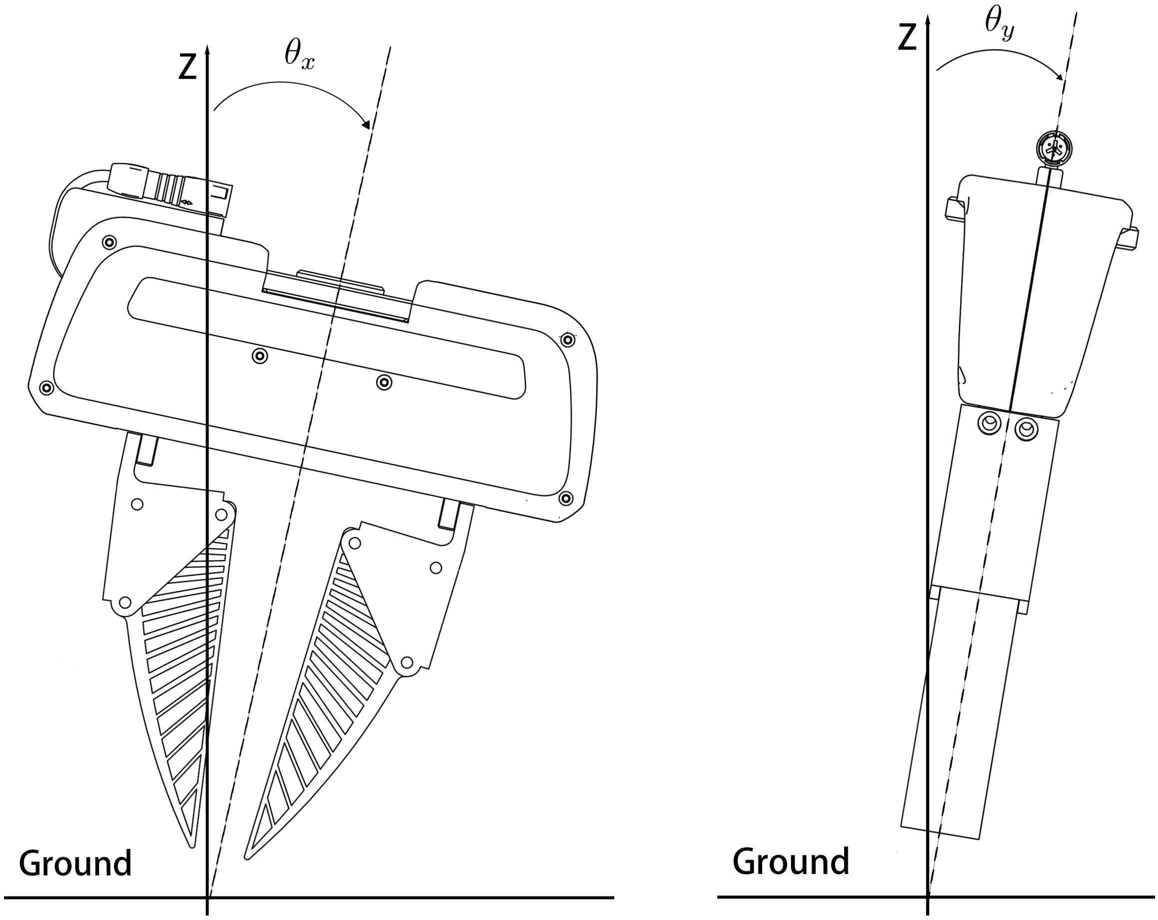
$$\mathbf{p}_{\mathcal{F}}^{*} = \mathbf{p}_{\mathcal{T}}^{\text{target}} - R \begin{bmatrix} 0 \\ 0 \\ z_{\text{TCP}} \end{bmatrix}. \tag{3.3}$$

This relation is implemented in the trial-control software: the target orientation $R = R_{\text{tilt}} \circ R_0$ is constructed at session start by composing the requested tilt rotation $R_{\text{tilt}} = R_y(\theta_y)R_x(\theta_x)$ (extrinsic, world frame) onto the base orientation $R_0 = R_x(180°)$, and the flange position target is updated accordingly. The orientation is then held constant through all subsequent descent and grasp phases, so the tip XY remains at (x_0, y_0) throughout the trial. Figure 3.3 illustrates the two runtime tilt angles. In the front view, θ_x rotates the gripper about the symmetric axis and both fingers move together; in the side view, θ_y rotates it about the gripping axis and the mirror-symmetric fingers overlap in projection. Both are shown here as a general rotation from the world z axis (“Ground”); the runtime sweep used in this study is $\theta_x, \theta_y \in \{-2.5°, 0°, +2.5°\}$. These arm rotation angles are a separate quantity from the seat angle α of Section 3.1.2. The seat angle is a fixed property of the gripper hardware, namely how the fingers meet the flange, while θ_x, θ_y describe how the arm is oriented relative to the surface. The two are reported independently throughout this thesis and are never summed.

For a $+2.5^\circ$ tilt about x on the `finray_26` variant ($z_{\mathrm{TCP}} = 94.911\,\mathrm{mm}$), the predicted flange offset is

$$\Delta y_{\mathcal{F}} = -z_{\mathrm{TCP}}\sin(2.5^\circ) \approx -4.14\,\mathrm{mm}. \tag{3.4}$$

Chapter 4 reports the empirical lateral offsets measured under load and the residuals from this geometric prediction, which we attribute to finger compliance under press.



(a) θ_x: rotation about the symmetric axis (front view)

(b) θ_y: rotation about the gripping axis (side view)

Figure 3.3: Runtime tilt angles θ_x (left) and θ_y (right), each shown as a rotation of the whole gripper body away from the world z axis (labelled) relative to the Ground line; 0° is the arm perpendicular to the surface.

3.5 Data Pipeline

3.5.1 Per-Trial Recording

Each trial directory `trial_NNN/` contains:

- `config.yaml`: snapshot of the variant config at trial time;
- `run_config.json`: depth, tilt, ground-truth source, speeds, gripper mode;
- `pose_wrench.csv`: high-rate (30 Hz) actual + commanded pose, RPY, wrench, gripper width per row, tagged with the current trial stage.

3.5.2 Per-Session Recording

Each session directory `depth_*/` contains in addition:

- `depth_summary.json`: header metadata + array of per-trial verdicts (`success`, `fail_reason`); operator notes;
- `ground_truth_used.json`: snapshot of the GT JSON loaded;
- `session_log_*.npz`: numpy archive of the 30 Hz background sample stream, with stage labels and tap indices.

3.5.3 Verdict Recording and Outcome Classification

Trial outcomes were classified into three categories rather than a binary success/failure. After each trial the operator was prompted to distinguish *success* (the gripper held the cloth throughout the lift stage) from *no-grasp* (the trial executed to completion and the stationary force sample was valid, but the cloth slipped from the gripper during or after the lift). Trials in which the FCI control daemon aborted the motion mid-trial were recorded as *failed trials* and excluded from all aggregate analyses; the two most common triggers were:

- the Franka safety reflex firing on excessive predicted joint force (`auto_failed_excess_force`), which typically occurred when the parallel-jaw variants were commanded to depths beyond their narrow force-limited working range;

- the Franka safety reflex firing on acceleration discontinuity between two consecutive Cartesian trajectory segments, which was observed intermittently during the chunked touch-off descent and mitigated by raising the trajectory-emission floor of the touch-off descent trajectories.

Both trigger classes left the trial's `success` field as `null` in the per-trial JSON; no-grasp trials, by contrast, carried `success = false` with a valid force record and were retained for engagement-force analysis.

3.6 Trial Procedure

A single trial proceeded as follows:

0. Cloth reset: the operator flattened the cloth on the test surface, smoothing out any wrinkles or folds left by the previous trial, restoring the flat-on-surface starting configuration;

1. The gripper opened to the Franka Hand or finray release width;

2. Bulk descent: a smooth multi-waypoint trajectory carried the gripper from the lifted pose down to a fine clearance (+5 mm above surface), filtered to exclude any upward leg, preventing direction reversals at zero-twist interior waypoints;

3. Settle: a 2 s pause at the fine-clearance pose;

4. Final approach: a cosine-S-curve descent carried the gripper from the fine clearance to the press target $z_{\mathrm{surf}} - d$, where d is the trial press depth (Figure 3.4);

5. Grasp: a force-controlled close using the Franka grasp action at 80 N or 20 N nominal, with ±100 mm ϵ to disable the position-acceptance check that would otherwise cause auto-release on a sub-spec final width;

6. Settle: a 1 s pause;

7. Stationary sample: 30 samples of $\mathbf{F}_{\mathrm{ext}}$ were captured over 1 s during the `grasp_stationary_sample` stage;

8. The gripper lifted to $\sim +80$ mm above surface for visual inspection;

9. The operator was prompted for a verdict;
10. The gripper opened and the rig was reset for the next trial.

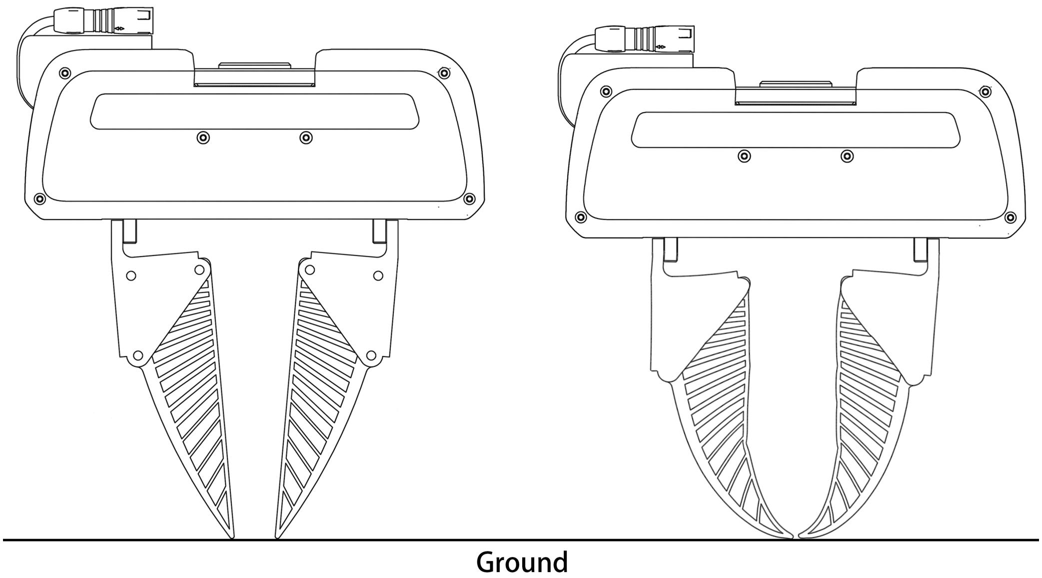


Figure 3.4: Physical effect of the press depth d on a finray variant. At first contact ($d = 0$, left) the blades are straight and only the tips touch the surface; pressed to $d > 0$ (right), the compliant blades buckle inward and wrap the surface, increasing the contact area. This wrap-around engagement underlies the depth-gated tilt robustness discussed in Chapter 6.

3.7 Statistical Treatment

3.7.1 Calibrated Depth Reference

All aggregate analyses are reported on a *calibrated* depth axis

$$d_{\text{cal}} = d - d_{\text{first success}}, \tag{3.5}$$

where $d_{\text{first success}}$ is the shallowest cell in which at least two-thirds of the valid completed trials succeed, the majority-success criterion. It is computed per trial series, one series being a variant–tilt-condition pair within a calibration epoch, and each series is zeroed at its own first majority-success cell. The reference is thus a directly measured physical event, the onset of reliable grasping, rather than the force-touch-off zero, which inherits cross-session bias drift and per-print geometry offsets (Section 3.2.2). Aggregate statistics and figures use only $d_{\mathrm{cal}} \in [0, 10]$ mm; cells outside this window remain in the record but are excluded from aggregates. The 10 mm bound was chosen because, at that width, every finray series' record extends at least to the window end, so every finray range in Chapter 5 is *right-censored* uniformly: the true working range is known only to be at least 10 mm, since the underlying record continues beyond the window but is excluded from the aggregate by convention, not because the gripper stopped working there. The bound also keeps every finray's peak–valley shape (Section 5.5) fully contained within the window. A wider window (e.g. 12 mm) was evaluated and rejected because one finray variant's shape curve extended only to 10.5 mm of usable support at that width, breaking the uniform-censoring property. The rigid parallel-jaw variants' working ranges are unaffected by the window choice, as both lie entirely inside $[0, 10]$ mm and their ranges are therefore reported as closed measurements, not censored. This convention was adopted as Bill 0007 of the project's governance record, enacted 2026-08-09 and revised through 2026-08-10 to per-series zeroing, a majority-success cell criterion, and the final 10 mm window, on the evidence that first-success alignment collapses the two prints of the 18 mm finger onto a common curve.

3.7.2 Per-Cell Statistics

For each (variant, depth, tilt) cell, the contact force is reported as

$$\bar{F}_z = \frac{1}{N} \sum_{i=1}^{N} F_z^{(i)}, \qquad \sigma_{F_z} = \sqrt{\frac{1}{N} \sum (F_z^{(i)} - \bar{F}_z)^2}, \tag{3.6}$$

where N is the total number of stationary samples pooled across all clean trials in the cell. Trials with a null verdict, i.e. those ended by an FCI crash, are excluded. Trial-to-trial variance is the dominant source of dispersion; the within-trial sample variance (~ 0.1 N) is small compared to it.

4

Experimental Design and Procedure

The empirical campaign was organised into two sequential phases on a single experimental apparatus. Phase 1 established the contact-force versus press-depth relationship for each gripper variant at nominal zero-tilt orientation. Phase 2 then perturbed the same measurement with $\pm 2.5^\circ$ tilts about the two orthogonal Cartesian axes. Together the two phases define a three-dimensional (depth, tilt-axis, tilt-sign) cell grid. The per-cell measurements of that grid constitute the data corpus reported in Chapter 5.

4.1 Study Design

The gripper roster comprised six variants, organised into two families.

- **Finray family**, four variants. The family comprises `finray_18`; `finray_18_model2`, a second print of the same CAD; `finray_26`, with a larger blade; and `finray_26_5deg`, which is `finray_26` on a 5°-inclined seat.

- **Parallel-jaw family**, two variants. These are `parallel_jaw_stock`, the bare rigid jaws, and `parallel_jaw_TPU`, the same jaws with a 2 mm TPU 90A compliant pad.

The same fabric specimen (single-layer cotton) rested on the same acrylic table surface throughout the campaign. The Franka FR3 arm held the gripper in a fixed base orientation $R_0 = R_x(180°)$ for Phase 1. For Phase 2, signed rotations $R_\theta = R_x(180°)\,R_y(\theta_y)\,R_x(\theta_x)$ were applied, with $(\theta_x, \theta_y) \in \{(\pm 2.5°, 0), (0, \pm 2.5°)\}$. These are the runtime arm tilts of Chapter 3, illustrated in Figure 3.3. They are distinct from the seat angle α of Figure 3.1.

The perturbation magnitudes were chosen to represent the placement uncertainty of a deployment that cannot afford precise calibration. A low-accuracy or coarsely mounted robot arm may plausibly carry an orientation error of a few degrees and a surface-height error of several millimetres. The $\pm 2.5°$ runtime tilt stands in for that orientation error. The depth sweep spans the corresponding height-error budget, and its 10 mm calibrated analysis window is defined in Chapter 3. Therefore, a gripper that tolerates the full tested range also tolerates a deployment in which neither the mounting angle nor the surface height is precisely known.

Table 4.1 lists the test conditions common to both phases.

Table 4.1: Test conditions of the experimental campaign. Symbols are defined in the Notation section; θ_x, θ_y are illustrated in Figure 3.3.

Variable	Definition	Values / precision
Press depth d	Fingertip driven below the detected surface	rigid: -1 to $+4$ mm at 0.25–0.5 mm steps finray: -2 to $+16$ mm at 0.5–1.0 mm steps
Tilt θ_x	Arm leaned about the symmetric axis	$-2.5°$, $0°$, $+2.5°$
Tilt θ_y	Arm leaned along the gripping direction	$-2.5°$, $0°$, $+2.5°$
Surface reference	Force touch-off at 3 N threshold	repeatable to ± 0.1 mm
Repeats	Trials per (variant, depth, tilt) cell	3

4.2 Depth Grid

The depth axis is defined relative to the per-session detected cloth surface via the `ground_truth_finder` touch-off procedure described in Chapter 3. The depth grid was chosen adaptively for each gripper family.

- For the rigid parallel jaws, depths were spaced at 0.25–0.5 mm intervals over −1 to +4 mm, close to the narrow force-limited working range.
- For the finrays, depths were spaced at 0.5–1.0 mm intervals over −2 to +16 mm, covering the four regimes (boundary, knee, valley, recovery) identified in Chapter 5.

Three trials were collected per (variant, depth, tilt) cell. Boundary depths were routinely retested when the within-cell standard deviation exceeded 1 N, up to a maximum of two additional trials per (depth, tilt) combination.

4.3 Tilt-Aware Pose Geometry: Verification

Before reporting force results, the implementation of the tilt-aware tool-centre-point (TCP) tracking equation was verified empirically. Table 4.2 shows the measured flange XY shift during the stationary grasp sample versus the geometric prediction for `finray_26` at four representative (depth, tilt) cells.

Table 4.2: Measured versus predicted flange-XY offset during the stationary grasp sample at four cells. Predicted offset uses $z_{\text{TCP}} = 94.911$ mm.

Cell	Tilt cmd	Predicted $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Measured $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Residual	Mean F_z
−1.0 mm	+2.5° y	$(+4.14, +0.00)$	$(+4.35, -0.03)$	0.21 mm	−8.00 N
−1.0 mm	−2.5° y	$(-4.14, +0.00)$	$(-4.02, -0.04)$	0.13 mm	−3.39 N
+10.0 mm	+2.5° y	$(+4.14, +0.00)$	$(+5.35, -0.28)$	1.24 mm	−36.26 N
+10.0 mm	−2.5° y	$(-4.14, +0.00)$	$(-2.93, -0.29)$	1.24 mm	−35.95 N

At boundary depth, the residual was $\leq$ 0.21 mm, which is approximately the lateral positioning noise of the arm under light load. At safe depth, both positive and negative tilts exhibited a +1.2 mm common-mode offset in the same $+x$ direction, regardless of tilt sign. This common-mode offset is interpreted as evidence of finger compliance under load. The $\sim 36\,\mathrm{N}$ axial press deflects the soft finray blade laterally by approximately 1.2 mm in a structurally preferred direction. The differential offset (between $+y$ and $-y$ tilts) remains correctly predicted by the geometric model in both depth conditions.

This verification confirms that the control-frame implementation of the tilt-aware TCP tracker is correct. Any asymmetries observed in the contact force therefore reflect physical gripper behaviour instead of control-system errors.

4.4 Rotation Tracking Verification

The tilt conditions assume that the Franka arm realises the commanded orientation faithfully. The 30 Hz session log records actual and commanded RPY at every sample, giving a direct measurement of the orientation tracking error

$$\delta(t) = \|\mathrm{RPY}_{\mathrm{actual}}(t) - \mathrm{RPY}_{\mathrm{cmd}}(t)\|_2. \tag{4.1}$$

Across five representative sessions, the mean steady-state δ was $0.05°$ with a worst-case $0.13°$. Transient excursions up to $\sim 5°$ occur exclusively within the `orientation_lock` stage at session start, before any descent or grasp. Such excursions are filtered out of trial-level analyses.

A $2.5°$ commanded tilt is therefore $50\times$ the noise floor of the arm's rotation tracking. The observed force differences cannot be attributed to arm rotation error.

4.5 Trial Protocol and Outcome Classification

Each trial followed the fixed stage sequence detailed in Section 3.6. The commanded pose was set above the target, and the arm descended to the pre-contact height. The gripper then contacted the cloth and pressed to the target depth, closed with its default close command, and lifted by a fixed clearance. The operator's held/not-held verdict was recorded at the end of the trial. A stationary-sample window of 1 s (30 samples) during the grasped-and-pressed stage provided the force measurement.

Trial outcomes were classified into three categories.

Success: the trial executed to completion, contact force was sampled cleanly during the stationary window, and the cloth remained gripped after the lift stage.

No-grasp: the trial executed to completion and contact force was sampled cleanly during the stationary window, but the cloth slipped from the fingers during or after the lift stage. The force measurement remains valid as an *engagement-force* characterisation.

Failed trial: the trial did not execute to completion. The controller either aborted with an excess-force safety event or the operator interrupted before the stationary window completed. The measurement is invalid and the trial is excluded from analyses.

Section 6.2 discusses why this three-way taxonomy was adopted in place of a binary success/failure classification.

4.6 Experiment 3: Inclined-Surface Demonstration (Pilot)

Experiments 1 and 2 characterise the contact phase on a flat, levelled surface, and each variant is calibrated against that surface. Experiment 3 tests the deployment consequence that those measurements predict. In this experiment the surface geometry departs from what the calibration assumed, while the calibration, the control policy and the sensing are left unchanged. The question is whether the compliant finger keeps grasping under that departure where the rigid jaw fails.

The planned protocol mirrors Experiments 1 and 2 in every respect except the surface and the outcome metric. The acrylic test surface is mounted on a wedge, `SURFACE_RISER`, whose CAD model specifies a top-surface incline of $\beta = 4.15°$. Measured from the model geometry, the top face normal is 4.1467° from vertical. The slope is aligned with the gripping direction, and the as-built fixture will be verified against the nominal value before the session. Each variant is then calibrated *blind* by the standard force touch-off of Section 3.2.2, exactly as if the surface were flat and level. No tilt compensation is applied, and the incline is treated as unknown to the calibration. Each trial presses to the calibrated touch-off depth plus 0.5 mm, which is the shallowest depth at which every variant grasped successfully on the flat surface. The outcome is recorded as a binary grasp verdict under the outcome taxonomy of

Section 4.5. Contact force is logged, although it is not a reported metric for this experiment.

The pilot covers all six gripper variants of Table 3.1, at three attempts each, giving 18 trials in total. From the flat-surface working ranges of Chapter 5, the registered prediction is that all four finray variants grasp successfully at every attempt, while both rigid variants fail or are marginal. A 4.15° surface-relative misalignment exceeds the largest runtime tilt at which the rigid jaws retained a usable window (2.5°). Under that tilt the window had already narrowed to between 0.5 and 2.5 mm. In addition, one `parallel_jaw_stock` tilt condition was reduced to a single successful depth, and two produced no success at all (Section 5.3).

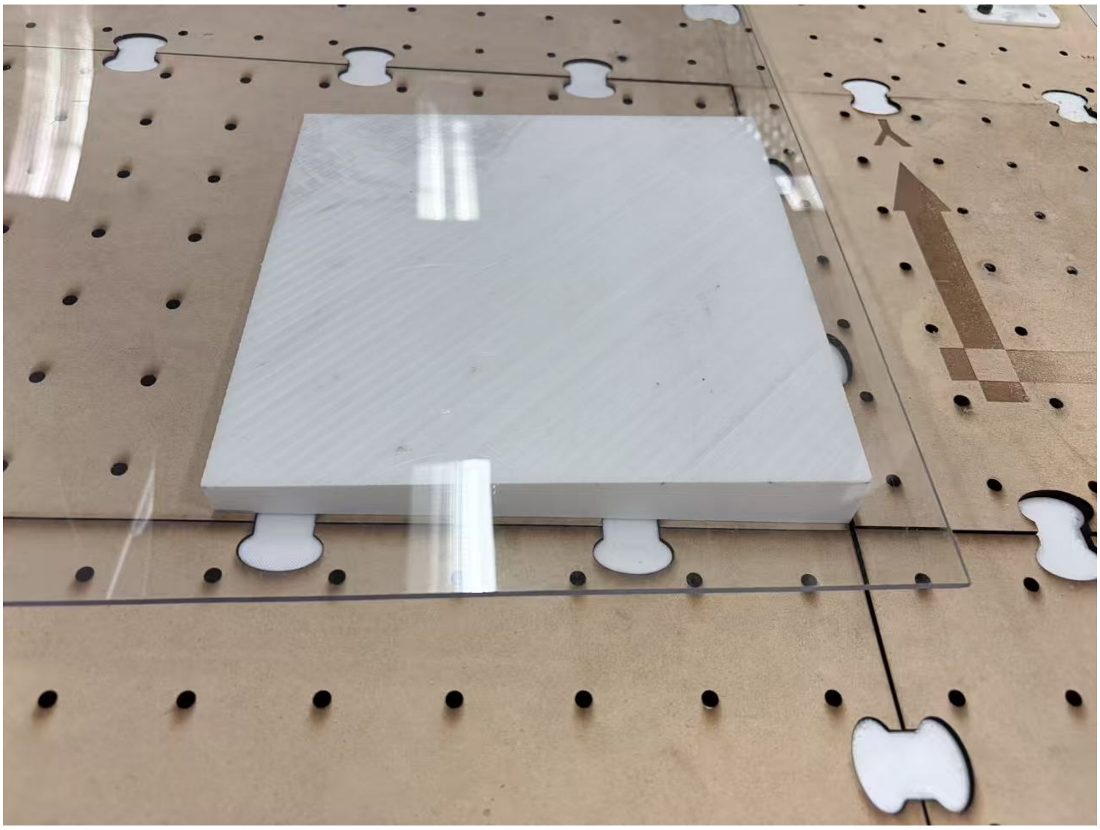


Figure 4.1: The `SURFACE_RISER` wedge (3D-printed, white) mounted under the acrylic test surface, on the optical breadboard used for Experiments 1 and 2. The arrow marks the gripper's local y axis (the gripping direction), along which the wedge slope is aligned.

[TODO: Experiment 3 is a placeholder: the protocol above is specified (Bill 0009, pending enactment) and the wedge fixture is built (Figure 4.1), but the pilot has not yet been run. Insert the outcome grid (variant $\times$ attempt, success/no-grasp/abort), the as-built incline angle if it differs from the CAD nominal $4.15°$, and a short results paragraph once the session is complete. If the pilot is not run before submission, delete this section and retain the inclined-surface demonstration as future work in Chapter 7.]

5

Results

This chapter reports the empirical findings across all six gripper variants, namely `finray_18`, `finray_18_model2`, `finray_26`, `finray_26_5deg`, `parallel_jaw_stock`, and `parallel_jaw_TPU`. The findings are organised around the primary measurement, the contact force F_z, at swept command depths and $\pm 2.5°$ tilts about the x and y axes. All values were F_z-bias-corrected using the per-session Franka F_{ext} zero-load reading. The corrected values were then pooled across trials with matching command depths and tilts unless otherwise noted. Interpretation is deferred to Chapter 6.

5.1 Print-to-Print Manufacturing Variability

Figure 5.1 compares two independent prints of the same `finray_18` CAD geometry on the calibrated depth axis. Each print was zeroed at its own first majority-success cell (Section 3.7.1). The two prints differed by a +4.6 N mean force offset, with the second print being the stiffer, and the RMSE over the 10 mm window was 5.4 N. Removing the constant offset left a residual RMSE of 2.9 N. This ≈ 5 N offset was adopted as the canonical same-session print-to-print variability, approximately 15% of the knee peak force.

Cross-session drift on the *same* print was measured by comparing `finray_18` data collected on separate days approximately three weeks apart. That drift was approximately 7–10 N at knee and knee-adjacent depths. Therefore, drift across sessions on a single specimen was larger than same-session print-to-print variability.

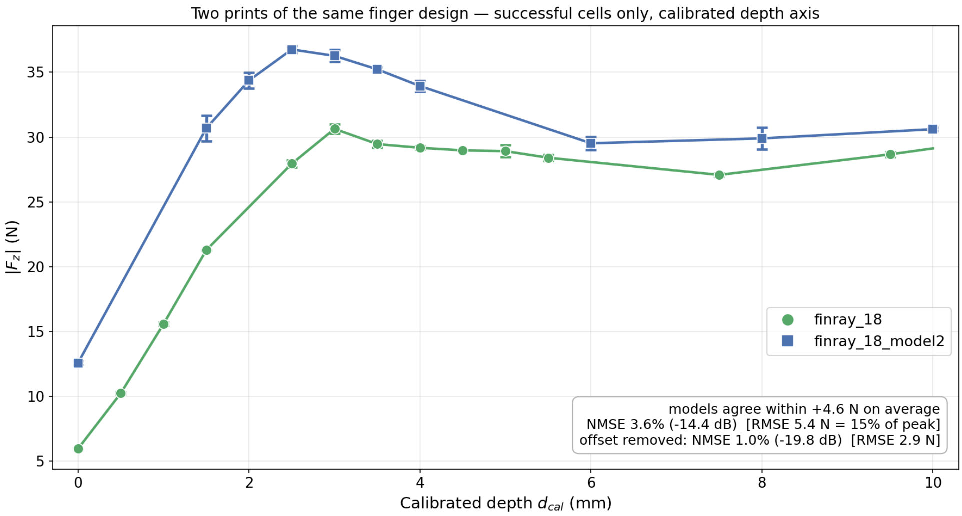


Figure 5.1: Contact force versus calibrated depth for two independently printed copies of `finray_18`, zero tilt, successful cells only. Error bars: one standard deviation across repeats; where no bar is visible, the spread is smaller than the marker.

The two prints' primary force-depth response was otherwise nearly identical (pairwise shape NMSE 0.41%, Table 5.2). Their secondary x-axis bias also shared the same depth-gated envelope as the other bias measurements in this work (Section 5.6), being largest near the boundary and relaxing toward zero at depth. The two prints differed only in where that shared envelope sat relative to zero. The bias of `finray_18` crossed zero, running negative at the boundary, briefly positive through the knee, and negative again at depth. However, the bias of `finray_18_model2` stayed negative throughout the full measured range. The gap between the two was itself depth-gated, at 2–3.5 N near the boundary and shrinking to < 1 N by $d = +10$ mm. The attribution of this offset is deferred to Section 6.1.

5.2 Working Range at Zero Tilt

Figure 5.2 displays the successful-grasp working range of each variant at zero tilt, on the calibrated depth axis of Section 3.7.1. The parallel-jaw variants exhibited closed windows of 1.5 mm (`parallel_jaw_stock`) and 2.5 mm (`parallel_jaw_TPU`). In contrast, all four finray variants spanned the full ≥ 10 mm analysis window. Their ranges were right-censored at the window end, and the underlying record extends beyond it.

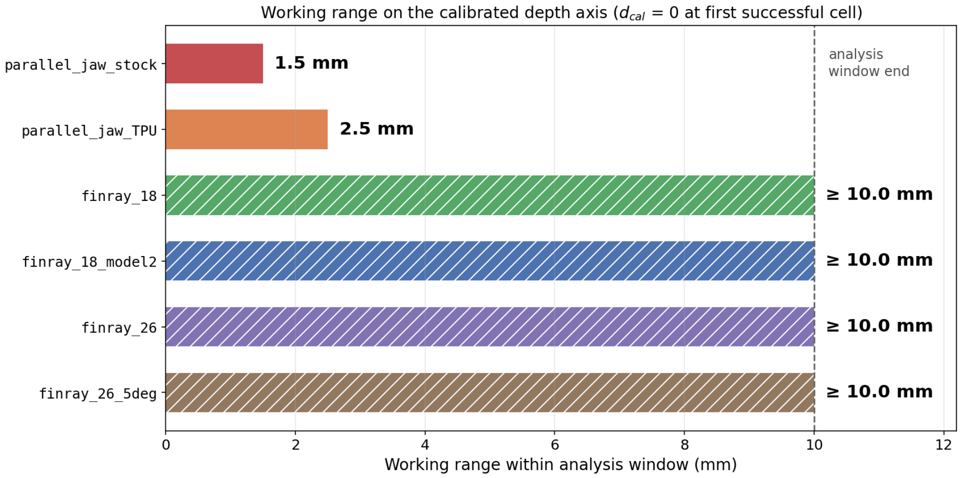


Figure 5.2: Successful-grasp working range on the calibrated depth axis, zero tilt. Hatched bars are right-censored at the 10 mm analysis-window end (Section 3.7.1).

5.3 Working Range Under Tilt

Figure 5.3 displays the same working ranges over every tested orientation. Extending the comparison in this way narrowed the parallel-jaw windows further, to 0.5–2.5 mm. One stock-jaw tilt condition was reduced to a single successful depth (0 mm span), and two others produced no successful grasp at any depth. Five orientations were tested per finray variant and four per parallel-jaw variant, whose $\theta_x = -2.5°$ condition is covered by mirror symmetry instead of by separate cells. Every finray variant retained its full ≥ 10 mm window under every orientation tested.

The parallel jaws auto-failed on the Franka excess-force safety monitor at depths only 1–1.5 mm beyond their successful-grasp window. That monitor triggers at -60 N

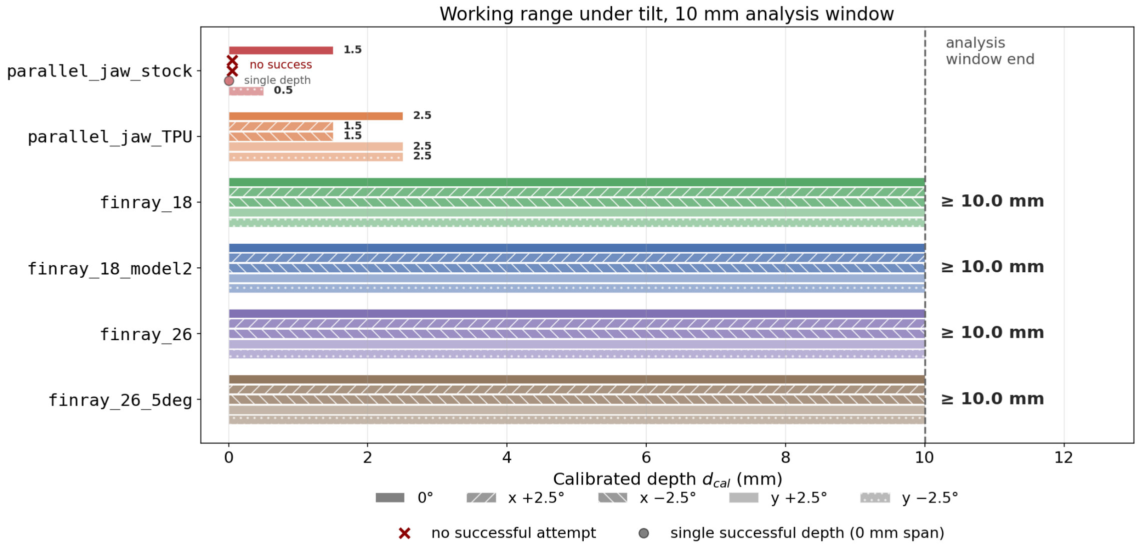


Figure 5.3: Successful-grasp working range on the calibrated depth axis, all tested orientations per variant (five for the finrays, four for the parallel jaws). ×: no successful attempt at any depth. Hatched bars are right-censored at the analysis-window end.

in F_z. However, the finrays never triggered the safety monitor within the tested −2 to +16 mm range on any variant.

5.4 Tilt-Aligned Response Comparison

Figure 5.4 displays the five tilt-condition curves of each finray variant together on the calibrated depth axis. Within each variant, the orientations tracked one another closely. The mean spread across the five orientations, averaged over the calibrated window, was 1.3–2.4 N depending on variant, which is small relative to the 31–53 N knee peaks reported in Table 5.1. The curves separated near the boundary and converged in the plateau, consistent with the depth-gated tilt robustness reported in Section 5.6.

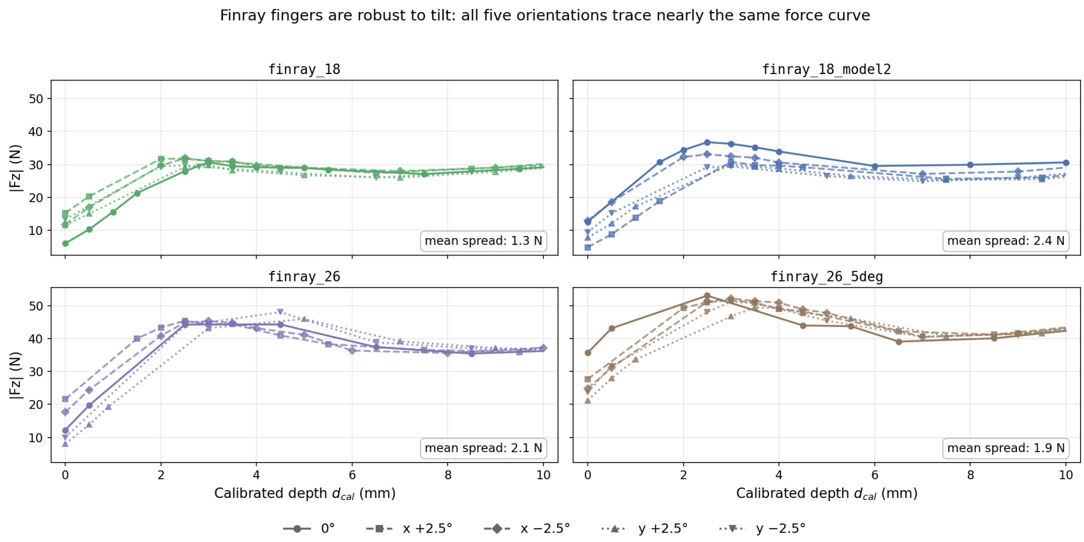


Figure 5.4: Contact force versus calibrated depth for each finray variant, all five tested orientations overlaid (0°, $x \pm 2.5°$, $y \pm 2.5°$). Within each panel the five curves nearly coincide.

5.5 One Shared Curve Shape Across the Finray Family

Under zero tilt, all four finray variants exhibited a stereotyped *bell-plus-valley* force-depth response with four regions. These were (i) a boundary regime where contact force rose steeply with press depth, (ii) a knee peak at calibrated depths of +2.5 to +4.5 mm, (iii) a valley trough near +6 to +8.5 mm, and (iv) a recovery regime toward the window end. Table 5.1 lists the key features for each variant. The four features lay within $\approx \pm 1.5$ mm of the same depth locations across all four variants, and the absolute force magnitudes scaled approximately with the frontal contact area of the finger.

Normalising each variant's curve by its own peak force and averaging over its five tested orientations makes this shared shape directly visible (Figure 5.5): all four finray variants collapse onto a single rise–peak–valley–recovery curve over the full $[0, 10]$ mm calibrated window. Table 5.2 quantifies the pairwise agreement between variants as a normalised mean-squared error (NMSE) and, in decibels, $10 \log_{10}(\text{NMSE})$; the mean

Table 5.1: Key features of the 0° force-depth curve for each finray variant, on the calibrated depth axis (Section 3.7.1). Peak $|F_z|$ is the maximum observed contact force in the knee region; valley $|F_z|$ is the local minimum between the knee peak and the deep-regime recovery. The finray_26 knee is a flat plateau spanning +2.5 to +4.5 mm; its listed peak depth is the maximum-force cell.

Variant	Knee peak $\|F_z\|$	Peak depth	Valley $\|F_z\|$	Valley depth
finray_18	30.6 N	+3.0 mm	27.1 N	+7.5 mm
finray_18_model2	36.8 N	+2.5 mm	29.5 N	+6.0 mm
finray_26	44.3 N	+4.5 mm	35.5 N	+8.5 mm
finray_26_5deg	53.1 N	+2.5 mm	39.1 N	+6.5 mm

pairwise NMSE was 0.58% (−22.4 dB), with the largest disagreement between any two variants at 0.79% (−21.0 dB).

Table 5.2: Pairwise shape agreement between finray variants over the full $[0, 10]$ mm calibrated window: NMSE (top), NMSE in decibels (middle), and RMSE as a percentage of peak (bottom, in brackets).

	finray_18	finray_18_model2	finray_26	finray_26_5deg
finray_18	—	0.41% (−23.9 dB) [5.5%]	0.69% (−21.6 dB) [7.2%]	0.79% (−21.0 dB) [7.8%]
finray_18_model2		—	0.13% (−28.8 dB) [3.1%]	0.70% (−21.6 dB) [7.1%]
finray_26			—	0.75% (−21.2 dB) [7.4%]
finray_26_5deg				—

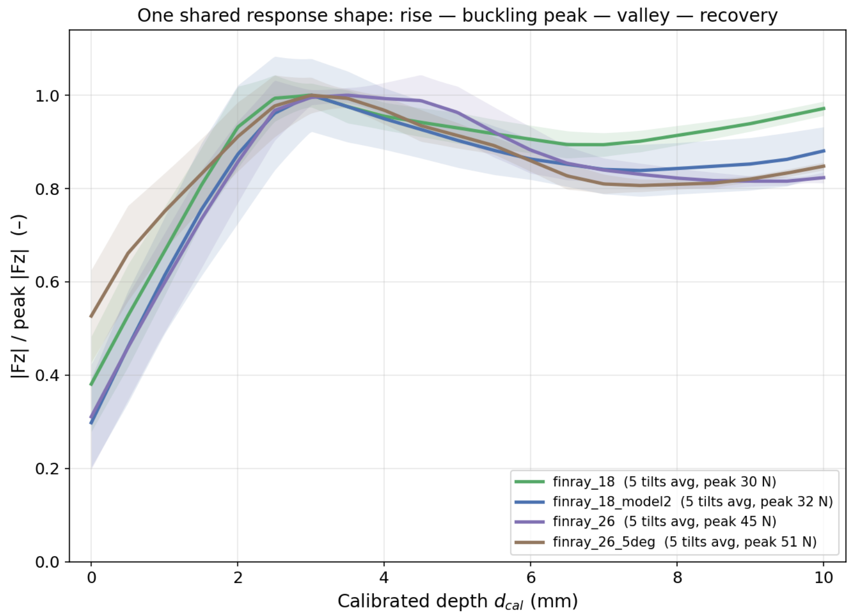


Figure 5.5: Peak-normalised $|F_z|$ versus calibrated depth d_{cal}, averaged over all five tested orientations per variant (shaded band: across-orientation spread). All four finray variants collapse onto one shared response shape.

5.6 Bipolar Tilt Bias on the Finray Family

A $\pm 2.5°$ tilt produced a signed bias $b(d, a) = |F_z^{+2.5°}(d, a)| - |F_z^{-2.5°}(d, a)|$. Depths in this section and the next are quoted on the raw command-depth axis d, the native axis of the per-cell bias record, rather than the calibrated axis d_{cal} used elsewhere in this chapter. The bias's depth trajectory itself followed a bipolar shape: a strong signed peak at boundary depths, a zero-crossing near the knee, a sign-reversed plateau through the valley, and depth-gated attenuation toward zero at deep press. The clearest example was the y-axis bias on `finray_26_5deg`, which had full coverage across 14 sign-flip pairs from -2 to $+16$ mm:

- Boundary peak: $b = +5.40 \pm 0.12$ N at $d = -1.5$ mm.
- Zero crossing: between $d = -0.5$ mm ($b = +2.00$ N) and $d = +1.5$ mm ($b = -1.35$ N).
- Knee plateau: $b = -1.57 \pm 0.21$ N at $d = +2.0$ mm, persisting to $d = +3.0$ mm.

- Depth-gated recovery: $|b| < 0.9\,\mathrm{N}$ at all depths beyond $d = +4\,\mathrm{mm}$.

The x-axis bias on `finray_26_5deg` had a similar bipolar shape but with opposite sign at the boundary ($b_x = -4.17 \pm 0.32\,\mathrm{N}$ at $d = -1.5\,\mathrm{mm}$).

5.7 Cross-Axis Sign Reversal at the Boundary

At boundary depths, the x-axis and y-axis biases had *opposite signs* on `finray_26_5deg` (Table 5.3).

Table 5.3: Bias on `finray_26_5deg` at boundary depths. x-axis and y-axis biases had opposite signs (sign product -1). Values are $|F_z^{+2.5°}| - |F_z^{-2.5°}|$ in newtons.

Depth	x-bias	y-bias	Sign product
$-1.5\,\mathrm{mm}$	-4.17 ± 0.32	$+5.40 \pm 0.12$	-1
$-1.0\,\mathrm{mm}$	-3.94 ± 0.18	$+4.27 \pm 0.07$	-1
$-0.5\,\mathrm{mm}$	-3.51 ± 0.39	$+2.00 \pm 0.24$	-1

The same sign-reversal pattern was present on `finray_18` at $d = -1.0\,\mathrm{mm}$ (x-bias $= -2.44 \pm 0.17\,\mathrm{N}$; y-bias $= +4.49 \pm 0.47\,\mathrm{N}$), demonstrating that the effect was not specific to the 5° mount condition.

In absolute terms, the boundary y-axis asymmetry corresponds to a $2.36\times$ force ratio between the two tilt signs. On `finray_26` at $d = -1.0\,\mathrm{mm}$, the mean stationary force was $8.00\,\mathrm{N}$ under $+2.5°$ against $3.39\,\mathrm{N}$ under $-2.5°$, while at $d = +10.0\,\mathrm{mm}$ the same pair differed by under 1%, at $36.26\,\mathrm{N}$ against $35.95\,\mathrm{N}$. It is worth noting that these values come from the four cells of the tilt-geometry verification (Table 4.2), which cover one variant on one axis, so the ratio should be read as an illustration of the depth gating reported in Section 5.6 and not as a separately established quantity.

5.8 Mount-Tilt Alignment: 5° Mount as a Depth Shift

After a two-parameter best-fit alignment, a single depth shift plus a single force offset, the 5°-mounted variant collapsed onto the flat-mounted variant with residual RMSE below $2\,\mathrm{N}$ on both tilt axes independently (Table 5.4).

Table 5.4: Best-fit alignment of `finray_26_5deg` onto `finray_26`. The two axes yielded independent estimates of the depth shift within 0.02 mm of each other.

Axis	Depth shift	Force offset	Alignment RMSE
x	−1.14 mm	+2.7 N	1.75 N
y	−1.12 mm	+3.1 N	1.58 N
mean	**−1.13 mm**	+2.9 N	1.66 N

6

Discussion

In this chapter, the results are interpreted in the order in which Chapter 5 reported them, beginning with the error structure of the measurements, followed by the working range at zero tilt, the effect of tilting, the similarity of the finray responses across orientation and depth, and finally the shared force-depth profile and the buckling mechanics behind it. The limitations bounding these claims close the chapter (Section 6.6).

6.1 Measurement and Manufacturing Error

The print-to-print comparison of Section 5.1 is also the error budget for the whole dataset, and its central finding is a noise hierarchy. Same-session print-to-print variability, a mean offset of 4.6 N with RMSE 5.4 N, proved *smaller* than the 7–10 N cross-session drift measured on the same print at knee depths. Therefore, the dominant nuisance variable in this dataset is the elapsed time between sessions. Candidate contributors include cloth conditioning, TPU viscoelastic recovery, and drift in the FCI external-wrench estimate, which is a model-based estimate instead of a load-cell reading [46]. Any cross-variant force claim with an effect size below the 5 N print-to-print offset is unlikely to reproduce across physical instances of the same CAD, and should be treated as tentative until replicated across prints. Likewise, any campaign that pools data across days must either report cross-day drift as a separate variance component or restrict its headline claims to same-session pairs; in this work, all bias values are same-session unless labelled otherwise, and cross-day comparisons are used only for shape consistency. Table 6.1 collects the error terms on a common scale.

Table 6.1: Error and variability terms of the campaign, ordered by magnitude. Percentages are relative to a nominal 30 N knee peak. The shape terms are computed on peak-normalised curves and so are insensitive to the force offsets above them.

Term	Magnitude	Relative	Source
Cross-session drift, same print	7–10 N	23–33%	Section 5.1
Print-to-print offset, same session	4.6 N	15%	Section 5.1
residual after offset removal	2.9 N	10%	Section 5.1
Boundary tilt bias, peak	5.4 N	18%	Section 5.6
Plateau tilt bias	< 0.9 N	$< 3\%$	Section 5.6
Across-orientation spread, calibrated	1.3–2.4 N	4–8%	Section 5.4
Within-cell retest trigger	1 N	3%	Section 4.2
Within-trial sample noise	~ 0.1 N	$< 1\%$	Section 3.7
Print-to-print shape (NMSE)	0.41%	−23.9 dB	Table 5.2
Cross-variant shape (NMSE, mean)	0.58%	−22.4 dB	Table 5.2
Cross-variant shape (NMSE, worst)	0.79%	−21.0 dB	Table 5.2

A central asymmetry of the dataset is visible in the table. Every force-magnitude term sits between 1 and 10 N, one to two orders of magnitude above the shape terms, which are below 1% normalised error. Therefore, absolute forces in this campaign are uncertain at the several-newton level, although the *shape* of the force-depth response is reproducible to a fraction of a percent. For this reason, shape agreement and depth locations are used in this work for the headline claims, and absolute force comparisons are treated as session-bounded.

The x-axis bias sign difference between the two prints, deferred here from Section 5.1, fits inside this error budget. Both prints' bias curves share the same depth-gated envelope, strongest at the boundary and fading with depth, and differ only in a shrinking, print-dependent offset (2–3.5 N near the boundary, < 1 N by $d = +10$ mm) that happens to carry `finray_18` across zero while `finray_18_model2` remains on one side throughout. The offset lives entirely on the secondary, off-axis force component and leaves the *shape* of the dominant F_z response essentially unchanged at 0.41% NMSE; the +4.6 N magnitude offset is a separate, primary-axis quantity, already counted in the noise hierarchy above. The data are consistent with more than one mechanism. Small print-instance differences in rib-wall thickness or infill density may manifest as a direction-dependent buckling offset. Alternatively, a small, direction-dependent bias in the F_z-dominant external-wrench estimate (Section 3.2.1) may be responsible, which would naturally be largest where the signal is weakest, at the boundary, and vanish where it is strongest, in the plateau. Distinguishing between them would require post-hoc CT scanning of the two prints or the independent force-sensor cross-check identified as future work (Chapter 7); until such a measurement is made, the finding should be read as a bound on secondary-axis force uncertainty, and whether the two prints respond differently to tilt still remains unclear.

In addition, the measurement design shapes what counts as valid data. The trial-outcome taxonomy of success, no-grasp, and failed trial (Section 4.5) is deliberately three-valued because the categories map to physically distinct regimes. Treating no-grasp trials as failures would delete clean force measurements from the boundary regime, truncating the observed curves and masking the boundary-tilt biases; treating failed trials as no-grasps would pollute the averages with readings from incomplete stationary windows, near-zero in interrupted trials and saturated in safety-reflex aborts. The three-way taxonomy is the minimum classification that separates "the trial produced valid measurement" from "the grasp was successful" from "the trial was invalid."

6.2 Working Range Without Tilt

At zero tilt, the two gripper classes occupy different regimes of the same axis. Closed windows of 1.5 and 2.5 mm were measured for the rigid jaws, against censored windows of at least 10 mm for every finray (Section 5.2). The failure behaviour differs as sharply as the window widths do. The parallel jaws auto-failed on the −60 N excess-force reflex within 1–1.5 mm of their window edge. However, the finrays never triggered the reflex anywhere in the tested −2 to +16 mm range. Therefore, compliance flattens the map from position error to force spike, and this property may matter more than the success rate itself in collaborative deployment.

For deployment, this means finray-based systems can run open-loop, with no depth-clipping controller and no safety-limit tuning. Parallel-jaw systems require an active depth governor with command precision well inside their 0.5–2.5 mm tilt-dependent window, since the jaws auto-failed within 1–1.5 mm beyond the window edge, leaving little margin for command error. Any cloth-handling system whose depth uncertainty exceeds roughly half the rigid-jaw window should default to a finray-class gripper. The no-grasp regime at the window's shallow edge is also where a grasp controller could in principle intervene productively. The pre-lift press force is measured and the depth is known, so a marginal engagement could be flagged before the lift begins, and the controller could then deepen the press or abort before the cloth is dropped outside the pick zone.

6.3 Effect of Tilting

Tilting the arm degraded the two classes asymmetrically. The rigid windows shrank to 0.5–2.5 mm, with one stock-jaw condition reduced to a single successful depth and two producing no success at all. However, every finray variant kept its full window at every orientation (Section 5.3). Within the finrays, the tilt effect is depth-gated. Section 5.6 reported a strong signed asymmetry at boundary depths that attenuates to under 0.9 N beyond $d = +4$ mm, where depths in that record are raw command depths. This bipolar trajectory tracks the contact-geometry transition documented in Figure 3.4. At boundary depths only the blade tips touch the surface; the contact patch is small, and a 2.5° orientation change redistributes which tip carries the load, so the asymmetric blade profile dominates the force response. Past the buckling knee, the blades wrap the surface and the contact becomes an extended line along each blade; the same orientation change then only slightly redistributes load along an already-engaged contact. The bias attenuates at raw $d = +4$ mm, about $+4.5$ mm calibrated for this series, just past the knee-peak region. Therefore, the quantity that confers tilt robustness is the wrap-around engagement of the blades, and press depth acts only through the engagement it produces. This is the mechanism behind the depth-gated tilt robustness identified as the principal empirical contribution of this work (Chapter 7). Although compliance does not remove tilt sensitivity, it relocates the sensitivity to the contact-boundary regime, which a sufficiently deep press avoids.

Comparing the raw-axis bias curves with the per-series calibrated view of Section 5.4 implies that the raw-axis asymmetry is, to first order, an orientation-dependent shift in engagement onset seen through the local slope of the bell curve, while the force law itself appears unchanged. This inference is not circular, because the per-series zero is defined by grasp success and is independent of any feature of the force curve, so the collapse of the calibrated curves is evidence that the success onset and the force-curve position shift together under tilt. The reading is also quantitatively self-consistent, since dividing the +5.40 N boundary bias by the boundary-region slope of 5–10 N/mm implies an onset shift of 0.5–1 mm, at or below the cell spacing. In addition, it organises the bipolar trajectory. The bias is large where the curve is steep, and it reverses sign as the curve tops out approaching the knee; the measured crossing between $d = -0.5$ and $+1.5$ mm precedes the raw-axis knee peak near $+2$ mm by about a millimetre, consistent with the knee flattening ahead of its peak cell. At depth the bias attenuates faster than the local slope alone would predict, which suggests that the onset shift itself collapses as the blades wrap. It is worth noting that the 1.3–2.4 N residual spreads set the floor above which bias structure can be resolved, so only the boundary magnitude and the sign reversal are established above that floor. In addition, the asymmetry remains operationally decisive whatever its mechanism, because a deployed system calibrates once, blind to its own tilt error, and therefore operates on the raw axis.

The sign structure of the asymmetries separates their origins. The y-axis bias is intrinsic to the finger. The gripper is mirror-symmetric about the xz plane (Section 3.2.1), so rotation about x, the symmetric axis, is sign-equivalent by construction. However, rotation about y, the gripping direction, is not sign-equivalent, and a $+2.5°$ rotation about the gripping axis loads the two fingers differently and produces a signed force difference. The x-axis bias should vanish by that same symmetry. However, its nonzero value, with sign opposite to the y-bias (Section 5.7), indicates an additional symmetry-breaking contribution, whose origin still remains unclear. The cloth may carry an anisotropic warp–weft axis misaligned with the gripper's mirror plane, coupling through the whole-hand contact into a signed force difference. Alternatively, the asymmetry may sit on the finger or fixture side after all. The tilt-geometry verification (Table 4.2) already established a sign-independent $+1.2\,\mathrm{mm}$ common-mode deflection in $+x$ under $\sim 36\,\mathrm{N}$ press, and a fixed structural lean of this kind would also produce a signed x-bias at the boundary, since a $+2.5°$ x-tilt deepens the preferred lean while a $-2.5°$ tilt opposes it. The reproducibility of the x-bias across two variants does not discriminate, because both share the same seat design, printer, and process. Rotating the cloth specimen 90° in the lab frame is the discriminating experiment. A cloth-side origin would flip the x-bias sign, while a finger-side origin would leave it unchanged, and the y-bias should be approximately unchanged in either case.

6.4 Similarity Across Orientation and Depth

Once each series is re-zeroed at its own first majority-success cell, the five orientation curves of every finray variant nearly coincide, with mean spreads of only 1.3–2.4 N against knee peaks of 31–53 N (Section 5.4). Orientation moves the engagement onset but barely perturbs the force law that follows it. Therefore, the per-series calibration of Section 3.7.1 is more than a convenience. Its success in collapsing the orientations, and the two prints, is itself evidence that engagement onset is the physically meaningful origin for this contact problem.

The seat-mount comparison extends the same conclusion to a fixed, built-in tilt. Raising the seat angle from the $+2.5°$ baseline to $+5.0°$ proved equivalent to a $-1.13\,\mathrm{mm}$ depth shift, axis-independent to within $0.02\,\mathrm{mm}$ (Table 5.4). Because the comparison variant already sits on the $+2.5°$ baseline seat (Section 3.1.2), the differential tilt is 2.5°; a differential tilt of a rigid finger of length L produces a projected vertical offset of $L\,[\sin(5°) - \sin(2.5°)] = 0.0436L$, and setting this equal to the observed shift gives $L_{\mathrm{eff}} \approx 26\,\mathrm{mm}$, near the mid-length of the $\sim 50\,\mathrm{mm}$ finger, consistent with load distributed along the finray's curved contact surface instead of being concentrated at the distal tip. The axis-independence rules out any mechanism in which the pre-tilt produces a different effective load-bearing length on the two axes. Therefore, the seat-angle increment acts as a strictly rigid translation of the operating point in depth, shifting the envelope by 1.13 mm for a 2.5° increment at the cost of a $\sim 3\,\mathrm{N}$ vertical force offset, without altering the underlying force-depth sensitivity.

6.5 One Shared Profile: Peak, Valley, Recovery, and Buckling

The strongest regularity in the dataset is that all four finray variants, spanning two thicknesses, two prints, and two seat angles, collapse onto a single peak-normalised force-depth curve with a mean pairwise NMSE of 0.58% (Section 5.5). The curve's shape is consistent with the finger operating in three sequential contact regimes. In the boundary regime, only the distal tip contacts the cloth and force rises roughly linearly as contact area grows. At the knee, the full blade section is engaged and the finger resists compression through its rib structure, producing the peak. Past the peak, the ribs partially buckle and the finger folds back on itself, so that contact area increases while effective stiffness drops, producing the valley. At recovery depths the finger has fully wrapped, and further penetration is resisted by structural compression through the finger base. The four features, boundary, knee peak, valley, and recovery, occur at the same calibrated depths to within about $\pm 1.5\,\mathrm{mm}$ across all variants, which indicates that the three-regime response is a geometric property of the finray shape and is insensitive to blade thickness or seat angle. Force magnitudes scale with contact area, although the transition depths do not.

Read as structural mechanics, the curve is a buckling signature. Although force would be expected to grow monotonically with press depth under rigid contact, it does not do so here. The boundary rise is elastic pre-buckling loading of the straight blades; the knee peak is the limit point at which the blades reach their tip-load (buckling) limit; the valley is the post-buckling softening branch, in which the buckled blades carry less force even as displacement continues, the load drop characteristic of an elastic structure past a limit point; and the recovery is post-buckling re-stiffening, in which the inward-buckled blades come into extended contact with the surface, as shown in Figure 3.4, and further load transfers into structural compression through the finger base. This reading also explains why the transition depths are set by geometry, although the force magnitudes are not. A buckling limit is a property of the blade's shape and material, so geometrically similar blades place the peak and valley at the same depths while the critical load scales with blade section. This corresponds to the scale-not-shape separation reported in Section 5.5. No published finray model covers this axial post-buckling regime (Chapter 2); the analytical model identified in future work (Chapter 7) would connect it to the continuum-mechanics literature on post-buckling contact.

In addition, one design implication follows from the profile. The valley may be the preferred operating setpoint. It retains the full tilt insensitivity of the post-buckling regime while applying 20 to 30 percent less force than the knee peak (Table 5.1), and this margin matters for delicate or loosely woven fabrics.

6.6 Limitations

Single cloth specimen and single surface. The reported working ranges and cloth-fixture asymmetries were measured on one cotton cloth specimen on one acrylic table surface. The absolute numbers are expected to shift with cloth thickness, weave direction, friction, and surface compliance. The cross-axis sign reversal discussed in Section 6.3 requires the 90°-rotated cloth control to separate the cloth contribution from the finger and fixture contribution.

Cross-session drift. As established in Section 6.1, cross-session drift on the same specimen is the dominant noise source in this dataset; cross-day claims are qualified throughout.

Tilt range censoring. No finray failure was observed at any tested orientation, so the finray tilt windows are censored in angle, just as the depth windows are censored at 10 mm. Therefore, the data establish a floor of 2.5° runtime tilt, with 5° of seat inclination also tolerated, although the envelope edge itself was not reached. The angle at which finray grasping degrades still remains unclear, and Experiment 3's 4.15° incline probes only one point beyond the tested range.

Parallel-jaw x-axis coverage. The parallel-jaw variants were not swept across the x-axis tilt sign; the parallel-jaw bias values are restricted to the y-axis. Whether the x-axis bipolar bias observed on the finrays has an analogue on the rigid grippers still remains unclear.

Boundary-depth sample sizes. Boundary-depth cells frequently have $n = 3$, and boundary cells routinely triggered the $> 1\,\mathrm{N}$ within-cell retest criterion (Section 4.2), driven by cloth-fold state variability that is not further characterised. Boundary values are valid measurements with correspondingly wide error bars.

Success rate as a binary rating. The success/no-grasp distinction is operator-assessed after the lift stage; a finer-grained grip-force-during-lift measurement would resolve marginal cases that the current binary rating collapses.

6.7 Code and Data Availability

All source code, raw trial data, ground-truth JSON files, and analysis scripts are archived in the project repository; the public repository URL and the commit hash against which this thesis was written will be stated here at final submission. **[TODO: Insert public GitHub URL and commit hash before final submission.]**

7

Conclusion and Future Work

7.1 Answers to the Research Questions

In this work, a controlled experimental evaluation of six gripper variants was presented on a single platform (Franka FR3), measuring stationary contact force and grasp outcome as a function of press depth and tilt angle over 840 analysed trials, out of 955 collected in total (Appendix A). The four research questions posed in Chapter 1 are revisited below.

1. *How does contact force vary with press depth for each finger design?* Every finray variant follows a single shared rise–peak–valley–recovery curve, with pairwise NMSE below 1% over the full 10 mm calibrated window. However, the rigid parallel jaws ramp steeply toward the platform's excess-force limit within roughly 2 mm of first contact.

2. *How much can the gripper be tilted before the grasp becomes unreliable?* The finrays grasp reliably across every orientation tested, $\pm 2.5°$ about both axes

plus a 5° seat mount, over the full ≥ 10 mm window. However, the rigid jaws never exceed 2.5 mm, and two stock-jaw tilt conditions produced no successful grasp at any depth.

3. *Does tilt tolerance depend on press depth?* Yes. The robustness proved to be depth-gated. Near the contact boundary, tilt strongly modulates both the contact force and the grasp success. However, past the buckling threshold the mean across-orientation force spread shrinks to 1.3–2.4 N, against knee peaks of 31–53 N. A direction-dependent asymmetry about the geometrically asymmetric axis was also observed at the boundary, although it is supported by a small number of cells and disappears at the safe operating depths.

4. *What is the compliant finger's response under compression?* Blade buckling. Force peaks at buckling onset, $\approx$ 2–3 mm past first successful grasp, dips as the blades buckle, then recovers. Although the finray geometry, that is, blade thickness, seat angle, and print instance, sets the force *scale*, the curve *shape* is left unchanged, with knee peaks spanning 31–53 N.

The depth-dependent tilt robustness identified here had not been quantitatively characterised in prior literature, to the author's knowledge, and constitutes the principal empirical contribution.

7.2 Implications

With the robot, control policy, cloth, and environment held identical, a finger swap alone buys at least a 4× wider press-depth tolerance than the best rigid configuration, ≥ 10 mm versus 2.5 mm on the padded jaws and roughly 7× versus the stock jaws, at roughly half the grip force. Under tilt the advantage widens further, to 20× or more against the stock jaws' surviving conditions; all of these ratios are lower bounds, because the finray ranges are right-censored at the analysis-window end. The application domains that motivated this work are laundry and textile handling, hospital bedding, assistive robotics, garment manufacturing, and soft-goods packaging. For all of them, this means the flat-cloth pick, the step that currently keeps such systems human-started, becomes accessible to a standard arm running an unchanged policy, before any investment in cameras, tactile sensing, or learned control. The design rule that follows is simple: press the compliant finger past its buckling peak, $\approx$ 2–3 mm beyond first contact, and the grasp becomes robust to both depth and tilt placement error.

7.3 Future Work

Independent force-sensor cross-check. All force measurements derive from the arm's joint-torque-based external-wrench estimate. Repeating a subset of the matrix with a dedicated force/torque sensor at the wrist, or with an instrumented surface, would bound the estimator's pose-dependent bias directly.

Task-level evaluation. Folding a cloth diagonally from a corner grasp, dropping a centre-grasped cloth onto a curved object, and dragging between positions under the same policy would test whether the contact-phase advantage measured here translates into end-to-end task success, including under learned policies.

Task-level precision comparison. Beyond success rates, a soft versus rigid comparison of task-level *precision*, meaning placement accuracy after fold or drop, would characterise the cost side of compliance.

Dynamic and high-rate behaviour. All trials here are quasi-static. The perturbation behaviour of compliant fingers under fast actuation and high-frequency control, where the compliance that absorbs placement error may instead cause oscillation or delayed settling, is uncharacterised.

Tilt-sensitivity model. An analytical model relating finger curvature, contact compliance, and tilt-induced contact-point migration would connect the present empirical findings to the broader continuum-mechanics literature on soft contact.

Multi-fabric study. All trials used a single cloth specimen. Replicating the matrix on fabrics of varying thickness, weave, and surface treatment would establish the applicability range of the depth–tilt-robustness curves identified here.

Closed-loop control. The boundary-depth regime, where finray directional asymmetry dominates, motivates a tilt-compensating controller using inline force feedback, validated against the open-loop boundary baseline.

Sim2real validation. A simulation of the finray finger in a compliant-contact physics engine could be calibrated against the empirical force–depth–tilt surface from this thesis, providing a digital twin for the gripper variants.

A

Detailed Coverage and Supplementary Data

This appendix records the final per-variant coverage of the experimental campaign. The table below was regenerated programmatically from the on-disk trial record (`experiments/cloth_grasp/`) on 2026-08-04, counting only active cells; staged, superseded, and backup cells were excluded. A *cell* is one (variant, depth, tilt) session directory; each cell contains between one and several trials.

A.1 Per-Variant Coverage Table

In addition to the conditions tabulated above, `finray_26` carries five exploratory cells at $\theta_x = +5.0°$ (depths -0.5 to $+10$ mm, recorded 2026-06-05) and one zero-tilt boundary-depth cell recorded within the Experiment 2 campaign; these were pilot probes of the larger-tilt regime and are not part of the primary $\pm 2.5°$ analysis.

Across both experiments the active record comprises 955 trials. Not all of these enter the analyses, and the reconciliation is as follows. Fifteen trials belong to the

Table A.1: Experiment 1 (depth sweep at zero commanded tilt): active cell counts, sampled depth range, and total trials per variant.

Variant	Cells	Depth range (mm)	Trials
finray_18	21	$[-2, +14]$	56
finray_18_model2	28	$[-1, +14]$	86
finray_26	18	$[-1, +22]$	52
finray_26_5deg	11	$[-1, +20]$	31
parallel_jaw_TPU	9	$[-1, +4]$	26
parallel_jaw_stock	12	$[-1, +4]$	29
Total	99		280

Table A.2: Experiment 2 (tilt sweeps): active cell counts per commanded tilt condition, with sampled depth range in mm in brackets, and total trials per variant. The parallel-jaw variants could only be swept over their narrow force-limited depth window, and their $\theta_x = -2.5°$ condition is covered by mirror symmetry of the jaw about the xz plane rather than by separate cells.

Variant	θ_x=+2.5°	θ_x=−2.5°	θ_y=+2.5°	θ_y=−2.5°	Trials
finray_18	12 $[-1, +12]$	13 $[-2, +12]$	11 $[-2, +12]$	10 $[-2, +12]$	137
finray_18_model2	12 $[-2, +12]$	12 $[-2, +12]$	12 $[-2, +12]$	11 $[-2, +12]$	141
finray_26	14 $[0, +16]$	14 $[0, +16]$	11 $[-2, +16]$	10 $[-2, +16]$	145
finray_26_5deg	14 $[-2, +16]$	16 $[-2, +16]$	14 $[-2, +16]$	14 $[-2, +16]$	168
parallel_jaw_TPU	6 $[0, +4]$	—	5 $[0, +4]$	3 $[0, +3]$	40
parallel_jaw_stock	4 $[+1, +2]$	—	3 $[+1, +2]$	3 $[+1, +2]$	26
Total					657

exploratory cells beyond ±2.5° described above, and are excluded because they fall outside the tilt range of the primary study. A further 100 trials belong to cells that were later repeated at the same variant, depth, and tilt, usually after a recalibration or an interrupted session, and only the most recent valid cell of each configuration is used. The remaining 840 trials constitute the analysed record quoted in the Abstract and in Chapter 7. Of these, 837 completed the full trial sequence, and 10 of the completed trials were relabelled during curation as documented in Section 3.5.3, which leaves 827 trials contributing to the force statistics. No-grasp outcomes are retained as valid data per the outcome taxonomy, while failed trials are excluded from aggregate force analyses.

List of Publications

No portion of this thesis has been published, presented, or submitted for publication at the time of writing. A manuscript covering the depth–tilt evaluation results (Chapters 5 and 6) is in preparation.

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